ISC Class XII Mathematics

Chapter 2

Inverse Trigonometric Functions

75 Solved Problems – Numericals & Proofs

Step-by-Step ISC Board Solutions

Topics Covered

Compiled Answer Key – Step-by-step ISC board-style solutions

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1	Find the principal values of the following:	1	Numerical	2.1
(i)	$\sin^{-1}(\frac{1}{\sqrt{2}})$	1	Numerical	2.1
(ii)	$\cos^{-1}(\frac{1}{2})$	1	Numerical	2.1
(iii)	$\tan^{-1}(\sqrt{3})$	1	Numerical	2.1
2	Find the principal values of the following:	1	Numerical	2.1
(i)	$\sec^{-1}(\frac{2}{\sqrt{3}})$	1	Numerical	2.1
(ii)	$\cot^{-1}(\sqrt{3})$	1	Numerical	2.1
(iii)	$\text{cosec}^{-1}(\sqrt{2})$	1	Numerical	2.1
3	Find the principal values of the following:	1	Numerical	2.1
(i)	$\cos^{-1}(-\frac{1}{2})$	1	Numerical	2.1
(ii)	$\tan^{-1}(-1)$	1	Numerical	2.1
(iii)	$\text{cosec}^{-1}(-\sqrt{2})$	1	Numerical	2.1
4	Find the principal values of the following:	1	Numerical	2.1
(i)	$\cot^{-1}(-\frac{1}{\sqrt{3}})$	1	Numerical	2.1
(ii)	$\sec^{-1}(-\frac{2}{\sqrt{3}})$	1	Numerical	2.1
(iii)	$\sin^{-1}(-1)$	1	Numerical	2.1
5	Evaluate the following:	2	Numerical	2.1
(i)	$\cos^{-1}(\cos\frac{4\pi}{3})$	2	Numerical	2.1

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(ii)	$\sin^{-1}(\sin\frac{4\pi}{3})$	2	Numerical	2.1
(iii)	$\tan^{-1}(\tan\frac{3\pi}{4})$	2	Numerical	2.1
6	Evaluate the following:	2	Numerical	2.1
(i)	$\cos^{-1}(\cos\frac{5\pi}{3})$	2	Numerical	2.1
(ii)	$\tan^{-1}(\tan\frac{9\pi}{8})$	2	Numerical	2.1
(iii)	$\sec^{-1}(\sec\frac{9\pi}{5})$	2	Numerical	2.1
7	Show that:	2	Numerical	2.1
(i)	$\tan^{-1}(\tan\frac{5\pi}{6}) \neq \frac{5\pi}{6}$, what is its value?	2	Numerical	2.1
(ii)	$\cos^{-1}(\cos(-\frac{\pi}{6})) \neq -\frac{\pi}{6}$, what is its value?	2	Numerical	2.1
(iii)	$\sin^{-1}(\sin\frac{5\pi}{3}) \neq \frac{5\pi}{3}$, what is its value?	2	Numerical	2.1
8	Find the domain of the following functions:	3	Numerical	2.1
(i)	$\sin^{-1}(1-x)$	3	Numerical	2.1
(ii)	$\sec^{-1}(2x-3)$	3	Numerical	2.1
(iii)	$\cos^{-1}x + \sin x$	3	Numerical	2.1
9	See parts below:	4	Numerical	2.1
(i)	Write the range of one branch of $\sin^{-1}x$, other than the principal branch.	4	Numerical	2.1
(ii)	Write the range of one branch of $\cos^{-1}x$, other than the principal branch.	4	Numerical	2.1
10	Find the maximum value of $(\cos^{-1}x)^2$.	5	Numerical	2.1
11	If $\cot^{-1}(\frac{3}{4}) = x$, find the value of $\cos x$.	6	Numerical	2.1
12	See parts below:	6	Numerical	2.1
(i)	If $\tan^{-1}(\frac{3}{4}) = x$, find the values of $\cos x$ and $\sin x$.	6	Numerical	2.1
(ii)	If $\cot^{-1}(-\frac{1}{7}) = x$, find the values of $\sin x$ and $\cos x$.	6	Numerical	2.1
(iii)	If $\tan^{-1}x = \sin^{-1}(\frac{1}{2})$, find the value of x.	6	Numerical	2.1
13	Evaluate the following:	7	Numerical	2.1
(i)	$\cot(\tan^{-1}\sqrt{3})$	7	Numerical	2.1
(ii)	$\sin(\frac{\pi}{6} - \sin^{-1}(-\frac{\sqrt{3}}{2}))$	7	Numerical	2.1
(iii)	$\cos(\cos^{-1}(-\frac{\sqrt{3}}{2}) + \frac{\pi}{6})$	7	Numerical	2.1
14	Prove the following:	7	Proof	2.1
(i)	$\sin^{-1}(-\frac{1}{2}) + \cos^{-1}(-\frac{\sqrt{3}}{2}) = \frac{2\pi}{3}$	7	Proof	2.1
(ii)	$\tan^{-1}(-1) + \cos^{-1}(-\frac{1}{\sqrt{2}}) = \frac{\pi}{2}$	7	Proof	2.1
15	Using principal values, find the values of:	7	Numerical	2.1
(i)	$\cos^{-1}(\frac{1}{2}) - 2\sin^{-1}(-\frac{1}{2})$	7	Numerical	2.1

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
(ii)	$\tan^{-1}(\sqrt{3}) - \cot^{-1}(-\sqrt{3})$	7	Numerical	2.1
(iii)	$\tan^{-1}(1) + \cos^{-1}(-\frac{1}{2})$	7	Numerical	2.1
(iv)	$\operatorname{cosec}^{-1}(-1) + \cot^{-1}(-\frac{1}{\sqrt{3}})$	7	Numerical	2.1
16	Find the values of the following:	8	Numerical	2.1
(i)	$\tan^{-1}(2\cos(2\sin^{-1}\frac{1}{2}))$	8	Numerical	2.1
(ii)	$\tan^{-1}(2\sin(2\cos^{-1}\frac{\sqrt{3}}{2}))$	8	Numerical	2.1
(iii)	$\sin^{-1}(\frac{1}{2}\sin(2\cot^{-1}(-1)))$	8	Numerical	2.1
(iv)	$\cot(\sin^{-1}(\cos(\tan^{-1}1)))$	8	Numerical	2.1
17	Find the value of:	2	Numerical	2.1
(i)	$\tan^{-1}(\tan\frac{5\pi}{6}) + \cos^{-1}(\cos\frac{13\pi}{6})$	2	Numerical	2.1
(ii)	$\tan^{-1}(-\frac{1}{\sqrt{3}}) + \cot^{-1}(\frac{1}{\sqrt{3}}) + \tan^{-1}(\sin(-\frac{\pi}{2}))\ \ldots$	2	Numerical	2.1
18	*(see parts below)*	7	Numerical	2.1
(i)	Evaluate: $\sin^{-1}(\sin\frac{3\pi}{4}) + \cos^{-1}(\cos\pi) + \tan^{-1}(1)$	7	Numerical	2.1
(ii)	Evaluate: ...	7	Numerical	2.1
(iii)	Evaluate: ...	7	Numerical	2.1
19	*(see parts below)*	9	Numerical	2.1
(i)	Evaluate: $\cos(\sin^{-1}(-\frac{3}{5}))$	9	Numerical	2.1
(ii)	Evaluate: $\operatorname{cosec}(\cos^{-1}(-\frac{12}{13}))$	9	Numerical	2.1
(iii)	Evaluate: $\sin(2\cot^{-1}\frac{1}{5})$	9	Numerical	2.1
20	Show that: $\tan(\frac{1}{2}\sin^{-1}\frac{3}{4}) = \frac{4-\sqrt{7}}{2}$	10	Proof	2.1

Exercise 2.2 – Properties and Simplification of Inverse Trigonometric Functions

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
1	Evaluate the following:	2	Numerical	2.2
(i)	$\sin^{-1}(\sin 2)$	2	Numerical	2.2
(ii)	$\sin^{-1}(\sin 5)$	2	Numerical	2.2
(iii)	$\tan^{-1}(\tan 4)$	2	Numerical	2.2
(iv)	$\cos^{-1}(\cos 10)$	2	Numerical	2.2
2	Find the values of:	7	Numerical	2.2
(i)	$\sin(\sin^{-1}x + \cos^{-1}x), \quad \lvert x\rvert \le 1$	7	Numerical	2.2
(ii)	$\cot(\tan^{-1}x + \cot^{-1}x), \quad x \in \mathbb{R}$	7	Numerical	2.2
3	Find the values of:	7	Numerical	2.2
(i)	$\csc(\sin^{-1}x + \cos^{-1}x), \quad \lvert x\rvert \le 1$	7	Numerical	2.2
(ii)	$\cos(\sec^{-1}x + \csc^{-1}x), \quad \lvert x\rvert \ge 1$	7	Numerical	2.2
4	Find the values of:	9	Numerical	2.2
(i)	$\tan^{-1}(\sin(-\frac{\pi}{2}))$	9	Numerical	2.2
(ii)	$\cos(\tan^{-1}\frac{3}{4})$	9	Numerical	2.2
(iii)	$\tan(\cos^{-1}\frac{8}{17})$	9	Numerical	2.2
5	Simplify the following:	11	Numerical	2.2
(i)	$\tan^{-1}(\frac{\sin x}{1+\cos x})$	11	Numerical	2.2

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
(ii)	$\tan^{-1}(\frac{x}{\sqrt{1-x^2}})$, $\quad |x| < 1$	11	Numerical	2.2
(iii)	$\sin^{-1}(\frac{x}{\sqrt{1+x^2}})$	11	Numerical	2.2
(iv)	$\csc^{-1}(\frac{\sqrt{1+x^2}}{x})$	11	Numerical	2.2
(v)	$\cot^{-1}(\frac{x}{\sqrt{1-x^2}})$, $\quad |x| < 1$	11	Numerical	2.2
(vi)	$\csc^{-1}(\frac{1}{\sqrt{1-x^2}})$, $\quad |x| < 1$	11	Numerical	2.2
6	Solve the following equations for x:	12	Numerical	2.2
(i)	$\tan^{-1} x = \sin^{-1}(\frac{1}{\sqrt{2}})$	12	Numerical	2.2
(ii)	$\sin^{-1} x = \cos^{-1}(\frac{\sqrt{3}}{2})$	12	Numerical	2.2
(iii)	$\cos^{-1} x = \sin^{-1}(-\frac{1}{2})$	12	Numerical	2.2
(iv)	$\cos(\sin^{-1} x) = \frac{1}{2}$	12	Numerical	2.2
(v)	$4\sin^{-1} x + \cos^{-1} x = \pi$	12	Numerical	2.2
(vi)	$\sin^{-1} x - \cos^{-1} x = \frac{\pi}{6}$	12	Numerical	2.2
(vii)	$5\tan^{-1} x + 3\cot^{-1} x = 2\pi$	12	Numerical	2.2
(viii)	$\tan^{-1} x^{-1} = \cot^{-1}(\frac{4}{x})$, $\quad x > 0$	12	Numerical	2.2
7	If $\sin(\sin^{-1}\frac{2}{3} + \cos^{-1} x) = 1$, then find the value of x.	12	Numerical	2.2
8	Find the values of:	13	Numerical	2.2
(i)	$\tan^{-1}\frac{1}{2} + \tan^{-1}\frac{1}{3}$	13	Numerical	2.2
(ii)	$\cot^{-1} 2 + \cot^{-1} 3$	13	Numerical	2.2
9	If $\tan^{-1} x - \tan^{-1} y = \frac{\pi}{4}$, $\quad xy > -1$, then find the value of ...	13	Numerical	2.2
10	See parts below:	7	Numerical	2.2
(i)	If $\tan^{-1} x = \frac{\pi}{10}$ for some $x \in \mathbb{R}$, find the value ...	7	Numerical	2.2
(ii)	If $\sin^{-1} x + \sin^{-1} y = \frac{\pi}{2}$, then find the value of ...	7	Numerical	2.2
(iii)	If $\sec^{-1} x + \sec^{-1} y = \frac{2\pi}{3}$, then find the value of ...	7	Numerical	2.2
11	If $\cos^{-1} x + \cos^{-1} y + \cos^{-1} z = 3\pi$, then find the value of ...	5	Numerical	2.2
12	Prove the following:	13	Proof	2.2
(i)	$\tan^{-1}(\frac{1+x}{1-x}) = \frac{\pi}{4} + \tan^{-1} x$, $\quad x < 1$	13	Proof	2.2
(ii)	$\tan^{-1} x + \cot^{-1}(x+1) = \tan^{-1}(x^2+x+1)$	13	Proof	2.2
(iii)	$\cos[\tan^{-1}\{\cot(\sin^{-1} x)\}] = x$	13	Proof	2.2
13	Prove the following (for suitable values of x and y):	13	Proof	2.2
(i)	$\tan^{-1}\sqrt{x} + \tan^{-1}\sqrt{y} = \tan^{-1}\frac{\sqrt{x}+\sqrt{y}}{1-\sqrt{xy}}$ ...	13	Proof	2.2
(ii)	$\tan^{-1}\frac{x+\sqrt{x}}{1-x^{3/2}} = \tan^{-1} x + \tan^{-1}\sqrt{x}$	13	Proof	2.2
14	Write the following functions in simplest form:	11	Numerical	2.2

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
(i)	$\tan^{-1}(\sqrt{\frac{1-\cos x}{1+\cos x}})$	11	Numerical	2.2
(ii)	$\tan^{-1}(\frac{\cos x-\sin x}{\cos x+\sin x})$	11	Numerical	2.2
(iii)	$\tan^{-1}(\frac{2\sqrt{x}}{1-x})$	11	Numerical	2.2
(iv)	$\sin^{-1}(\sqrt{\frac{x}{1+x}})$	11	Numerical	2.2
(v)	$\tan^{-1}(\sqrt{\frac{1-x}{1+x}})$	11	Numerical	2.2
(vi)	$\cos^{-1}(1-2x^2)$	11	Numerical	2.2
(vii)	$\sec^{-1}(\frac{1}{2x^2-1})$	11	Numerical	2.2
(viii)	$\tan^{-1}(\frac{x}{\sqrt{a^2-x^2}})$	11	Numerical	2.2
15	Prove that: $\sin(\cot^{-1}(\cos(\tan^{-1}x))) = \sqrt{\frac{x^2+1}{x^2+2}}$	8	Proof	2.2
16	Find the value of ...	13	Numerical	2.2
17	Prove the following:	13	Proof	2.2
(i)	$\tan^{-1}\frac{1}{2}+\tan^{-1}\frac{2}{11}=\tan^{-1}\frac{3}{4}$	13	Proof	2.2
(ii)	$\tan^{-1}\frac{1}{7}+\tan^{-1}\frac{1}{13}=\tan^{-1}\frac{2}{9}$	13	Proof	2.2
(iii)	$\tan^{-1}2-\tan^{-1}1=\tan^{-1}\frac{1}{3}$	13	Proof	2.2
(iv)	$\tan^{-1}1+\tan^{-1}\frac{1}{2}+\tan^{-1}\frac{1}{3}=\frac{\pi}{2}$	13	Proof	2.2
(v)	Prove that $2\tan^{-1}\frac{1}{3}+\tan^{-1}\frac{1}{7}=\frac{\pi}{4}$.	13	Proof	2.2
(vi)	Prove that ...	13	Proof	2.2
(vii)	Prove that $\cot^{-1}1+\cot^{-1}2+\cot^{-1}3=\frac{\pi}{2}$.	13	Proof	2.2
(viii)	Prove that $\tan^{-1}2+\tan^{-1}3=\frac{3\pi}{4}$.	13	Proof	2.2
18	If ...	13	Proof	2.2
19	*(see parts below)*	13	Proof	2.2
(i)	Prove that $\sin^{-1}\frac{1}{\sqrt{5}}+\sin^{-1}\frac{2}{\sqrt{5}}=\frac{\pi}{2}$.	13	Proof	2.2
(ii)	Prove that $\cos^{-1}\frac{4}{5}+\cos^{-1}\frac{12}{13}=\cos^{-1}\frac{33}{65}$.	13	Proof	2.2
(iii)	Prove that $\cos^{-1}\frac{3}{5}+\sin^{-1}\frac{12}{13}=\sin^{-1}\frac{56}{65}$.	13	Proof	2.2
(iv)	Prove that $\tan^{-1}\frac{1}{3}+\sec^{-1}\frac{\sqrt{5}}{2}=\frac{\pi}{4}$.	13	Proof	2.2
(v)	Prove that $\cos^{-1}\frac{4}{5}+\tan^{-1}\frac{3}{5}=\tan^{-1}\frac{27}{11}$.	13	Proof	2.2
(vi)	Prove that $\cos^{-1}\frac{5}{\sqrt{41}}+\cot^{-1}\frac{4}{5}=\frac{\pi}{2}$.	13	Proof	2.2
(vii)	Prove that ...	13	Proof	2.2
(viii)	Prove that $\sin^{-1}\frac{8}{17}+\cos^{-1}\frac{4}{5}=\cot^{-1}\frac{36}{77}$.	13	Proof	2.2
(ix)	Prove that ...	13	Proof	2.2
(x)	Prove that ...	13	Proof	2.2
20	If $\cos^{-1}(\frac{x}{a})+\cos^{-1}(\frac{y}{b})=\alpha$, prove that ...	13	Proof	2.2
21	*(see parts below)*	9	Numerical	2.2
(i)	Evaluate $\tan(2\tan^{-1}\frac{1}{2}-\cot^{-1}3)$.	9	Numerical	2.2
(ii)	Evaluate $\tan(2\tan^{-1}\frac{1}{5}-\frac{\pi}{4})$.	9	Numerical	2.2

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
22	*(see parts below)*	9	Numerical	2.2
(i)	Find the value of $\tan(\cos^{-1}\frac{3}{5}+\tan^{-1}\frac{1}{4})$.	9	Numerical	2.2
(ii)	Find the value of $\cos(\sin^{-1}\frac{3}{5}+\sin^{-1}\frac{5}{13})$.	9	Numerical	2.2
(iii)	Find the value of $\cot(\cos^{-1}\frac{4}{5}+\sin^{-1}\frac{2}{\sqrt{13}})$.	9	Numerical	2.2
(iv)	Find the value of $\sin(2\tan^{-1}\frac{2}{3})+\cos(\tan^{-1}\sqrt{3})$.	9	Numerical	2.2
23	*(see parts below)*	12	Numerical	2.2
(i)	Solve the equation for x: ...	12	Numerical	2.2
(ii)	Solve the equation for x: $\tan^{-1}4x+\tan^{-1}6x=\frac{\pi}{4}$.	12	Numerical	2.2
(iii)	Solve the equation for x: $\tan^{-1}(x+1)+\tan^{-1}(x-1)=\tan^{-1}\frac{6}{17}$.	12	Numerical	2.2
(iv)	Solve the equation for x: ...	12	Numerical	2.2
(v)	Solve the equation for x: ...	12	Numerical	2.2
(vi)	Solve the equation for x: ...	12	Numerical	2.2
(vii)	Solve the equation for x: $\cos(\sin^{-1}x)=\frac{1}{9}$.	12	Numerical	2.2
(viii)	Solve the equation for x: $\cos(2\sin^{-1}x)=\frac{1}{9}$.	12	Numerical	2.2
(ix)	Solve the equation for x: $\cos^{-1}(\sin(\cos^{-1}x))=\frac{\pi}{6}$.	12	Numerical	2.2
(x)	Solve the equation for x: $\sin^{-1}(\cos(\sin^{-1}x))=\frac{\pi}{3}$.	12	Numerical	2.2
(xi)	Solve the equation for x: ...	12	Numerical	2.2
(xii)	Solve the equation for x: ...	12	Numerical	2.2

Formula & Standard Results Sheet

I. Domain, Range and Basic Identities (Ex 2.1)

Domains and Principal Value Ranges

$$\sin^{-1}x : \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \qquad \cos^{-1}x : [0, \pi]$$
$$\tan^{-1}x : \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \qquad \cot^{-1}x : (0, \pi)$$
$$\sec^{-1}x : [0, \pi] \setminus \left\{\frac{\pi}{2}\right\} \quad \operatorname{cosec}^{-1}x : \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \setminus \{0\}$$

All domains: $\sin^{-1}, \cos^{-1} : [-1, 1]$; $\tan^{-1}, \cot^{-1} : \mathbb{R}$; $\sec^{-1}, \operatorname{cosec}^{-1} : |x| \geq 1$.

Property I – Inverse-of-Function Identities

$$\sin(\sin^{-1}x) = x,\ |x| \leq 1 \quad \cos(\cos^{-1}x) = x,\ |x| \leq 1 \quad \tan(\tan^{-1}x) = x,\ x \in \mathbb{R}$$

Holds whenever the inverse function is defined at x.

Property II – Function-of-Inverse Identities

$$\sin^{-1}(\sin\theta) = \theta,\ \theta \in \left[-\tfrac{\pi}{2}, \tfrac{\pi}{2}\right] \quad \cos^{-1}(\cos\theta) = \theta,\ \theta \in [0, \pi] \quad \tan^{-1}(\tan\theta) = \theta,\ \theta \in \left(-\tfrac{\pi}{2}, \tfrac{\pi}{2}\right)$$

Valid *only* inside the stated principal-value interval; outside it, reduce θ into range first.

Property III – Reciprocal Identities

$$\sin^{-1}\left(\frac{1}{x}\right) = \operatorname{cosec}^{-1}x,\ |x| \geq 1 \qquad \cos^{-1}\left(\frac{1}{x}\right) = \sec^{-1}x,\ |x| \geq 1 \qquad \tan^{-1}\left(\frac{1}{x}\right) = \cot^{-1}x,\ x > 0$$

For $x < 0$, $\tan^{-1}\left(\frac{1}{x}\right) = -\pi + \cot^{-1}x$.

Property IV – Negative-Argument (Odd/Even) Identities

$$\sin^{-1}(-x) = -\sin^{-1}x \qquad \tan^{-1}(-x) = -\tan^{-1}x \qquad \operatorname{cosec}^{-1}(-x) = -\operatorname{cosec}^{-1}x$$

$\sin^{-1}, \tan^{-1}, \operatorname{cosec}^{-1}$ are odd functions.

Property IV – Negative-Argument (Supplementary) Identities

$$\cos^{-1}(-x) = \pi - \cos^{-1} x \qquad \cot^{-1}(-x) = \pi - \cot^{-1} x \qquad \sec^{-1}(-x) = \pi - \sec^{-1} x$$

$\cos^{-1}, \cot^{-1}, \sec^{-1}$ satisfy this supplementary rule instead of being odd.

Property V – Complementary-Angle Identities

$$\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}, \ |x| \le 1 \qquad \tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}, \ x \in \mathbb{R} \qquad \sec^{-1} x + \operatorname{cosec}^{-1} x = \frac{\pi}{2}, \ |x| \ge 1$$

The two inverse functions in each pair always sum to $\pi/2$ on their common domain.

II. Addition, Subtraction and Double-Angle Formulas (Ex 2.2)

Property VI(a) – Sum Formula for Tangent Inverse

$$\tan^{-1} x + \tan^{-1} y = \tan^{-1}\left(\frac{x+y}{1-xy}\right), \quad xy < 1$$

If $xy > 1$ with $x, y > 0$: add π; if $xy > 1$ with $x, y < 0$: subtract π.

Property VI(b) – Difference Formula for Tangent Inverse

$$\tan^{-1} x - \tan^{-1} y = \tan^{-1}\left(\frac{x-y}{1+xy}\right), \quad xy > -1$$

Used to combine or split two arctangent terms.

Property VII – Sum/Difference Formulas for Sine and Cosine Inverse

$$\sin^{-1} x + \sin^{-1} y = \sin^{-1}\left(x\sqrt{1-y^2} + y\sqrt{1-x^2}\right)$$

$$\cos^{-1} x + \cos^{-1} y = \cos^{-1}\left(xy - \sqrt{1-x^2}\sqrt{1-y^2}\right)$$

Valid for suitable x, y so that the resulting angle stays within the principal range; otherwise subtract from π or add π as needed.

Property VIII – Double-Angle (2x) Formulas

$$2\tan^{-1} x = \sin^{-1}\left(\frac{2x}{1+x^2}\right), \ |x| \le 1 \qquad 2\tan^{-1} x = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right), \ x \ge 0 \qquad 2\tan^{-1} x =$$

$$\tan^{-1}\left(\frac{2x}{1-x^2}\right), \ -1 < x < 1$$

These come from substituting $x = \tan\theta$ into $\sin 2\theta$, $\cos 2\theta$, $\tan 2\theta$.

Triple-Tangent Identity

$$\tan(A+B+C) = \frac{\tan A + \tan B + \tan C - \tan A \tan B \tan C}{1 - \tan A \tan B - \tan B \tan C - \tan C \tan A}$$

Used when three inverse-tangent terms are added and their sum is a known angle (e.g. π).

III. Conversion Techniques (Ex 2.1 & 2.2)

Right-Triangle Conversion Method

$$\text{If } \theta = \sin^{-1}\left(\frac{p}{h}\right), \text{ construct a right triangle with}$$

$$\text{opposite} = p, \ \text{hypotenuse} = h, \ \text{adjacent} = \sqrt{h^2 - p^2}$$

The standard technique for converting one inverse trig ratio into another (e.g. $\sin^{-1} \to \tan^{-1}$) by reading off sides of the constructed triangle.

Weierstrass (Half-Angle) Substitution

$$t = \tan\frac{\theta}{2} \ \Rightarrow \ \sin\theta = \frac{2t}{1+t^2}, \quad \cos\theta = \frac{1-t^2}{1+t^2}, \quad \tan\theta = \frac{2t}{1-t^2}$$

Useful for simplifying expressions of the form $\tan^{-1}(f(\sin\theta, \cos\theta))$.

Half-Angle Trigonometric Identities

$$1 - \cos x = 2\sin^2\frac{x}{2} \qquad 1 + \cos x = 2\cos^2\frac{x}{2} \qquad \sin x = 2\sin\frac{x}{2}\cos\frac{x}{2}$$

Frequently used to simplify $\sqrt{\frac{1-\cos x}{1+\cos x}}$-type expressions to $\tan\frac{x}{2}$ before applying $\tan^{-1}$.

2.1 – Principal Values, Domain and Range of Inverse Trigonometric Functions

Question 1

Find the principal values of the following:

(i) $\sin^{-1}\left(\frac{1}{\sqrt{2}}\right)$

(ii) $\cos^{-1}\left(\frac{1}{2}\right)$

(iii) $\tan^{-1}(\sqrt{3})$

TYPE 1 **Principal Values of Inverse Trigonometric Functions**

To Find: The principal values of $\sin^{-1}\left(\frac{1}{\sqrt{2}}\right)$, $\cos^{-1}\left(\frac{1}{2}\right)$, and $\tan^{-1}(\sqrt{3})$

Concept / Formula Used: $\sin^{-1} x$ $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ $\cos^{-1} x$ $[0, \pi]$ $\tan^{-1} x$ $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

Solution:

(i) Let $y = \sin^{-1}\left(\frac{1}{\sqrt{2}}\right)$.

$$\sin y = \frac{1}{\sqrt{2}}$$

Since $\sin\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$ and $\frac{\pi}{4} \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$:

$$y = \frac{\pi}{4}$$

(ii) Let $y = \cos^{-1}\left(\frac{1}{2}\right)$.

$$\cos y = \frac{1}{2}$$

Since $\cos\left(\frac{\pi}{3}\right) = \frac{1}{2}$ and $\frac{\pi}{3} \in [0, \pi]$:

$$y = \frac{\pi}{3}$$

(iii) Let $y = \tan^{-1}(\sqrt{3})$.

$$\tan y = \sqrt{3}$$

Since $\tan\left(\frac{\pi}{3}\right) = \sqrt{3}$ and $\frac{\pi}{3} \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$:

$$y = \frac{\pi}{3}$$

Final Answer

(i) $\sin^{-1}\left(\frac{1}{\sqrt{2}}\right) = \frac{\pi}{4}$

(ii) $\cos^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{3}$

(iii) $\tan^{-1}(\sqrt{3}) = \frac{\pi}{3}$

Question 2

Find the principal values of the following:

(i) $\sec^{-1}\left(\frac{2}{\sqrt{3}}\right)$

(ii) $\cot^{-1}(\sqrt{3})$

(iii) $\text{cosec}^{-1}(\sqrt{2})$

TYPE 1 Principal Values of Inverse Trigonometric Functions

To Find: The principal values of $\sec^{-1}\left(\frac{2}{\sqrt{3}}\right)$, $\cot^{-1}(\sqrt{3})$, and $\text{cosec}^{-1}(\sqrt{2})$

Concept / Formula Used: $\sec^{-1} x \in [0, \pi] \setminus \left\{\frac{\pi}{2}\right\}$ $\cot^{-1} x \in (0, \pi)$ $\text{cosec}^{-1} x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \setminus \{0\}$

Solution:

(i) Let $y = \sec^{-1}\left(\frac{2}{\sqrt{3}}\right)$.

$$\sec y = \frac{2}{\sqrt{3}}$$

Since $\sec\left(\frac{\pi}{6}\right) = \frac{2}{\sqrt{3}}$ and $\frac{\pi}{6} \in [0, \pi] \setminus \left\{\frac{\pi}{2}\right\}$:

$$y = \frac{\pi}{6}$$

(ii) Let $y = \cot^{-1}(\sqrt{3})$.

$$\cot y = \sqrt{3}$$

Since $\cot\left(\frac{\pi}{6}\right) = \sqrt{3}$ and $\frac{\pi}{6} \in (0, \pi)$:

$$y = \frac{\pi}{6}$$

(iii) Let $y = \text{cosec}^{-1}(\sqrt{2})$.

$$\text{cosec}\, y = \sqrt{2}$$

Since $\operatorname{cosec}\left(\frac{\pi}{4}\right) = \sqrt{2}$ and $\frac{\pi}{4} \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \setminus \{0\}$:

$$y = \frac{\pi}{4}$$

Final Answer

(i) $\sec^{-1}\left(\frac{2}{\sqrt{3}}\right) = \frac{\pi}{6}$

(ii) $\cot^{-1}(\sqrt{3}) = \frac{\pi}{6}$

(iii) $\operatorname{cosec}^{-1}(\sqrt{2}) = \frac{\pi}{4}$

Question 3

Find the principal values of the following:

(i) $\cos^{-1}\left(-\frac{1}{2}\right)$

(ii) $\tan^{-1}(-1)$

(iii) $\operatorname{cosec}^{-1}(-\sqrt{2})$

TYPE 1 Principal Values of Inverse Trigonometric Functions

To Find: The principal values of $\cos^{-1}\left(-\frac{1}{2}\right)$, $\tan^{-1}(-1)$, and $\operatorname{cosec}^{-1}(-\sqrt{2})$

Concept / Formula Used: $\cos^{-1}(-x) = \pi - \cos^{-1} x$ $\tan^{-1}(-x) = -\tan^{-1} x$ $\operatorname{cosec}^{-1}(-x) = -\operatorname{cosec}^{-1} x$

Solution:

(i) Let $y = \cos^{-1}\left(-\frac{1}{2}\right)$. Using the identity $\cos^{-1}(-x) = \pi - \cos^{-1} x$:

$$\begin{aligned}\cos^{-1}\left(-\frac{1}{2}\right) &= \pi - \cos^{-1}\left(\frac{1}{2}\right)\\ &= \pi - \frac{\pi}{3}\\ &= \frac{2\pi}{3}\end{aligned}$$

Since $\frac{2\pi}{3} \in [0, \pi]$, this is the principal value.

(ii) Let $y = \tan^{-1}(-1)$. Using the identity $\tan^{-1}(-x) = -\tan^{-1} x$:

$$\begin{aligned}\tan^{-1}(-1) &= -\tan^{-1}(1)\\ &= -\frac{\pi}{4}\end{aligned}$$

Since $-\frac{\pi}{4} \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, this is the principal value.

(iii) Let $y = \operatorname{cosec}^{-1}(-\sqrt{2})$. Using the identity $\operatorname{cosec}^{-1}(-x) = -\operatorname{cosec}^{-1} x$:

$$\begin{aligned}\operatorname{cosec}^{-1}(-\sqrt{2}) &= -\operatorname{cosec}^{-1}(\sqrt{2}) \\ &= -\frac{\pi}{4}\end{aligned}$$

Since $-\frac{\pi}{4} \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \setminus \{0\}$, this is the principal value.

Final Answer

(i) $\cos^{-1}\left(-\frac{1}{2}\right) = \frac{2\pi}{3}$

(ii) $\tan^{-1}(-1) = -\frac{\pi}{4}$

(iii) $\operatorname{cosec}^{-1}(-\sqrt{2}) = -\frac{\pi}{4}$

Common Mistake: Writing $\cos^{-1}\left(-\frac{1}{2}\right) = -\frac{\pi}{3}$ is a common error. For $\cos^{-1}(-x)$, the range is $[0, \pi]$, so it must evaluate to $\pi - \cos^{-1} x = \frac{2\pi}{3}$.

Question 4

Find the principal values of the following:

(i) $\cot^{-1}\left(-\frac{1}{\sqrt{3}}\right)$

(ii) $\sec^{-1}\left(-\frac{2}{\sqrt{3}}\right)$

(iii) $\sin^{-1}(-1)$

TYPE 1 Principal Values of Inverse Trigonometric Functions

To Find: The principal values of $\cot^{-1}\left(-\frac{1}{\sqrt{3}}\right)$, $\sec^{-1}\left(-\frac{2}{\sqrt{3}}\right)$, and $\sin^{-1}(-1)$

Concept / Formula Used: $\cot^{-1}(-x) = \pi - \cot^{-1} x$ $\quad \sec^{-1}(-x) = \pi - \sec^{-1} x$ $\quad \sin^{-1}(-x) = -\sin^{-1} x$

Solution:

(i) Using the identity $\cot^{-1}(-x) = \pi - \cot^{-1} x$:

$$\begin{aligned}\cot^{-1}\left(-\frac{1}{\sqrt{3}}\right) &= \pi - \cot^{-1}\left(\frac{1}{\sqrt{3}}\right) \\ &= \pi - \frac{\pi}{3} \\ &= \frac{2\pi}{3}\end{aligned}$$

Since $\frac{2\pi}{3} \in (0, \pi)$, this is the principal value.

(ii) Using the identity $\sec^{-1}(-x) = \pi - \sec^{-1} x$:

$$\begin{aligned}\sec^{-1}\left(-\frac{2}{\sqrt{3}}\right) &= \pi - \sec^{-1}\left(\frac{2}{\sqrt{3}}\right)\\ &= \pi - \frac{\pi}{6}\\ &= \frac{5\pi}{6}\end{aligned}$$

Since $\frac{5\pi}{6} \in [0, \pi] \setminus \left\{\frac{\pi}{2}\right\}$, this is the principal value.

(iii) Using the identity $\sin^{-1}(-x) = -\sin^{-1} x$:

$$\begin{aligned}\sin^{-1}(-1) &= -\sin^{-1}(1)\\ &= -\frac{\pi}{2}\end{aligned}$$

Since $-\frac{\pi}{2} \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, this is the principal value.

Final Answer

(i) $\cot^{-1}\left(-\frac{1}{\sqrt{3}}\right) = \frac{2\pi}{3}$

(ii) $\sec^{-1}\left(-\frac{2}{\sqrt{3}}\right) = \frac{5\pi}{6}$

(iii) $\sin^{-1}(-1) = -\frac{\pi}{2}$

Question 5

Evaluate the following:

(i) $\cos^{-1}\left(\cos\frac{4\pi}{3}\right)$

(ii) $\sin^{-1}\left(\sin\frac{4\pi}{3}\right)$

(iii) $\tan^{-1}\left(\tan\frac{3\pi}{4}\right)$

TYPE 2 **Evaluating Composite Inverse Trigonometric Functions**

To Find: The values of the given composite inverse trigonometric expressions

Concept / Formula Used: $f^{-1}(f(x)) = x \qquad x \qquad f^{-1}$

Solution:

(i) Since $\frac{4\pi}{3} \notin [0, \pi]$, we reduce the angle:

$$\begin{aligned}\cos\left(\frac{4\pi}{3}\right) &= \cos\left(2\pi - \frac{2\pi}{3}\right) \\ &= \cos\left(\frac{2\pi}{3}\right)\end{aligned}$$

Since $\frac{2\pi}{3} \in [0, \pi]$:

$$\begin{aligned}\cos^{-1}\left(\cos\frac{4\pi}{3}\right) &= \cos^{-1}\left(\cos\frac{2\pi}{3}\right) \\ &= \frac{2\pi}{3}\end{aligned}$$

(ii) Since $\frac{4\pi}{3} \notin \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, we rewrite the angle:

$$\begin{aligned}\sin\left(\frac{4\pi}{3}\right) &= \sin\left(\pi + \frac{\pi}{3}\right) \\ &= -\sin\left(\frac{\pi}{3}\right) \\ &= \sin\left(-\frac{\pi}{3}\right)\end{aligned}$$

Since $-\frac{\pi}{3} \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$:

$$\begin{aligned}\sin^{-1}\left(\sin\frac{4\pi}{3}\right) &= \sin^{-1}\left(\sin\left(-\frac{\pi}{3}\right)\right) \\ &= -\frac{\pi}{3}\end{aligned}$$

(iii) Since $\frac{3\pi}{4} \notin \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, we rewrite the angle:

$$\begin{aligned}\tan\left(\frac{3\pi}{4}\right) &= \tan\left(\pi - \frac{\pi}{4}\right) \\ &= -\tan\left(\frac{\pi}{4}\right) \\ &= \tan\left(-\frac{\pi}{4}\right)\end{aligned}$$

Since $-\frac{\pi}{4} \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$:

$$\begin{aligned}\tan^{-1}\left(\tan\frac{3\pi}{4}\right) &= \tan^{-1}\left(\tan\left(-\frac{\pi}{4}\right)\right) \\ &= -\frac{\pi}{4}\end{aligned}$$

Final Answer

(i) $\cos^{-1}\left(\cos\dfrac{4\pi}{3}\right) = \dfrac{2\pi}{3}$

(ii) $\sin^{-1}\left(\sin\dfrac{4\pi}{3}\right) = -\dfrac{\pi}{3}$

(iii) $\tan^{-1}\left(\tan\dfrac{3\pi}{4}\right) = -\dfrac{\pi}{4}$

Common Mistake: Blindly writing $\cos^{-1}(\cos\theta) = \theta$ without checking if $\theta \in [0, \pi]$ is an extremely common error. Always reduce θ to the principal domain first.

Question 6

Evaluate the following:

(i) $\cos^{-1}\left(\cos\dfrac{5\pi}{3}\right)$

(ii) $\tan^{-1}\left(\tan\dfrac{9\pi}{8}\right)$

(iii) $\sec^{-1}\left(\sec\dfrac{9\pi}{5}\right)$

TYPE 2 **Evaluating Composite Inverse Trigonometric Functions**

To Find: The values of the given composite inverse trigonometric expressions

Concept / Formula Used: $f^{-1}(f(x)) = x \qquad x$

Solution:

(i) Since $\dfrac{5\pi}{3} \notin [0, \pi]$:

$$\begin{aligned}\cos\left(\frac{5\pi}{3}\right) &= \cos\left(2\pi - \frac{\pi}{3}\right) \\ &= \cos\left(\frac{\pi}{3}\right)\end{aligned}$$

Since $\dfrac{\pi}{3} \in [0, \pi]$:

$$\begin{aligned}\cos^{-1}\left(\cos\frac{5\pi}{3}\right) &= \cos^{-1}\left(\cos\frac{\pi}{3}\right) \\ &= \frac{\pi}{3}\end{aligned}$$

(ii) Since $\dfrac{9\pi}{8} \notin \left(-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right)$:

$$\begin{aligned}\tan\left(\frac{9\pi}{8}\right) &= \tan\left(\pi + \frac{\pi}{8}\right)\\ &= \tan\left(\frac{\pi}{8}\right)\end{aligned}$$

Since $\dfrac{\pi}{8} \in \left(-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right)$:

$$\begin{aligned}\tan^{-1}\left(\tan\frac{9\pi}{8}\right) &= \tan^{-1}\left(\tan\frac{\pi}{8}\right)\\ &= \frac{\pi}{8}\end{aligned}$$

(iii) Since $\dfrac{9\pi}{5} \notin [0, \pi] \setminus \left\{\dfrac{\pi}{2}\right\}$:

$$\begin{aligned}\sec\left(\frac{9\pi}{5}\right) &= \sec\left(2\pi - \frac{\pi}{5}\right)\\ &= \sec\left(\frac{\pi}{5}\right)\end{aligned}$$

Since $\dfrac{\pi}{5} \in [0, \pi] \setminus \left\{\dfrac{\pi}{2}\right\}$:

$$\begin{aligned}\sec^{-1}\left(\sec\frac{9\pi}{5}\right) &= \sec^{-1}\left(\sec\frac{\pi}{5}\right)\\ &= \frac{\pi}{5}\end{aligned}$$

Final Answer

(i) $\cos^{-1}\left(\cos\dfrac{5\pi}{3}\right) = \dfrac{\pi}{3}$

(ii) $\tan^{-1}\left(\tan\dfrac{9\pi}{8}\right) = \dfrac{\pi}{8}$

(iii) $\sec^{-1}\left(\sec\dfrac{9\pi}{5}\right) = \dfrac{\pi}{5}$

Question 7

Show that:

(i) $\tan^{-1}\left(\tan\dfrac{5\pi}{6}\right) \neq \dfrac{5\pi}{6}$, what is its value?

(ii) $\cos^{-1}\left(\cos\left(-\dfrac{\pi}{6}\right)\right) \neq -\dfrac{\pi}{6}$, what is its value?

(iii) $\sin^{-1}\left(\sin\dfrac{5\pi}{3}\right) \neq \dfrac{5\pi}{3}$, what is its value?

TYPE 2 **Evaluating Composite Inverse Trigonometric Functions**

To Prove: $\tan^{-1}\left(\tan\frac{5\pi}{6}\right) \neq \frac{5\pi}{6}$, $\cos^{-1}\left(\cos\left(-\frac{\pi}{6}\right)\right) \neq -\frac{\pi}{6}$, and $\sin^{-1}\left(\sin\frac{5\pi}{3}\right) \neq \frac{5\pi}{3}$, and evaluate each.

Concept / Formula Used: $f^{-1}(f(x)) = x \qquad x \qquad f^{-1}(x)$

Solution:

(i) The principal value branch of $\tan^{-1} x$ is $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

Since $\frac{5\pi}{6} \notin \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, it follows that $\tan^{-1}\left(\tan\frac{5\pi}{6}\right) \neq \frac{5\pi}{6}$.

Now, rewriting the trigonometric argument:

$$\begin{aligned}\tan\left(\frac{5\pi}{6}\right) &= \tan\left(\pi - \frac{\pi}{6}\right)\\ &= -\tan\left(\frac{\pi}{6}\right)\\ &= \tan\left(-\frac{\pi}{6}\right)\end{aligned}$$

Since $-\frac{\pi}{6} \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$:

$$\tan^{-1}\left(\tan\frac{5\pi}{6}\right) = \tan^{-1}\left(\tan\left(-\frac{\pi}{6}\right)\right) = -\frac{\pi}{6}$$

(ii) The principal value branch of $\cos^{-1} x$ is $[0, \pi]$.

Since $-\frac{\pi}{6} \notin [0, \pi]$, it follows that $\cos^{-1}\left(\cos\left(-\frac{\pi}{6}\right)\right) \neq -\frac{\pi}{6}$.

Now, using $\cos(-\theta) = \cos\theta$:

$$\cos\left(-\frac{\pi}{6}\right) = \cos\left(\frac{\pi}{6}\right)$$

Since $\frac{\pi}{6} \in [0, \pi]$:

$$\cos^{-1}\left(\cos\left(-\frac{\pi}{6}\right)\right) = \cos^{-1}\left(\cos\frac{\pi}{6}\right) = \frac{\pi}{6}$$

(iii) The principal value branch of $\sin^{-1} x$ is $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

Since $\frac{5\pi}{3} \notin \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, it follows that $\sin^{-1}\left(\sin\frac{5\pi}{3}\right) \neq \frac{5\pi}{3}$.

Now, rewriting the trigonometric argument:

$$\begin{aligned}\sin\left(\frac{5\pi}{3}\right) &= \sin\left(2\pi - \frac{\pi}{3}\right)\\ &= -\sin\left(\frac{\pi}{3}\right)\\ &= \sin\left(-\frac{\pi}{3}\right)\end{aligned}$$

Since $-\frac{\pi}{3} \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$:

$$\sin^{-1}\left(\sin\frac{5\pi}{3}\right) = \sin^{-1}\left(\sin\left(-\frac{\pi}{3}\right)\right) = -\frac{\pi}{3}$$

Final Answer

(i) Value $= -\frac{\pi}{6}$

(ii) Value $= \frac{\pi}{6}$

(iii) Value $= -\frac{\pi}{3}$

Question 8

Find the domain of the following functions:

(i) $\sin^{-1}(1-x)$
(ii) $\sec^{-1}(2x-3)$
(iii) $\cos^{-1}x + \sin x$

TYPE 3 **Domain of Inverse Trigonometric Functions**

★ Likely Exam Question

To Find: The domain of definition of each function

Concept / Formula Used: $\sin^{-1}u$ $\cos^{-1}u$ $-1 \leq u \leq 1$ $\sec^{-1}u$ $u \leq -1$ $u \geq 1$

Solution:

(i) For $\sin^{-1}(1-x)$ to be defined:

$$-1 \leq 1 - x \leq 1$$

Subtracting 1 throughout:

$$-1 - 1 \leq -x \leq 1 - 1$$
$$-2 \leq -x \leq 0$$

Multiplying by -1 (which reverses the inequality signs):

$$0 \leq x \leq 2$$

Thus, Domain $= [0, 2]$.

(ii) For $\sec^{-1}(2x-3)$ to be defined:

$$\begin{aligned} 2x - 3 \leq -1 \quad &\text{or} \quad 2x - 3 \geq 1 \\ 2x \leq 2 \quad &\text{or} \quad 2x \geq 4 \\ x \leq 1 \quad &\text{or} \quad x \geq 2 \end{aligned}$$

Thus, Domain $= (-\infty, 1] \cup [2, \infty)$.

(iii) For $f(x) = \cos^{-1} x + \sin x$:

- Domain of $\cos^{-1} x$ is $[-1, 1]$.
- Domain of $\sin x$ is $\mathbb{R} = (-\infty, \infty)$.

The domain of the sum is the intersection of the two domains:

$$\text{Domain} = [-1, 1] \cap (-\infty, \infty) = [-1, 1]$$

Final Answer

(i) Domain $= [0, 2]$

(ii) Domain $= (-\infty, 1] \cup [2, \infty)$

(iii) Domain $= [-1, 1]$

Question 9

(i) Write the range of one branch of $\sin^{-1} x$, other than the principal branch.
(ii) Write the range of one branch of $\cos^{-1} x$, other than the principal branch.

TYPE 4 Ranges of Inverse Trigonometric Branches

To Find: One non-principal branch range for $\sin^{-1} x$ and $\cos^{-1} x$

Concept / Formula Used: $\sin^{-1} x$ $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ $\cos^{-1} x$ $[0, \pi]$ π

Solution:

(i) For $\sin^{-1} x$, the principal value branch is $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.
Other branches on which $\sin x$ is bijective include:

$$\left[\frac{\pi}{2}, \frac{3\pi}{2}\right] \quad \text{or} \quad \left[-\frac{3\pi}{2}, -\frac{\pi}{2}\right]$$

Taking one such branch: $\left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$.

(ii) For $\cos^{-1} x$, the principal value branch is $[0, \pi]$.
Other branches on which $\cos x$ is bijective include:

$$[\pi, 2\pi] \quad \text{or} \quad [-\pi, 0]$$

Taking one such branch: $[\pi, 2\pi]$.

Final Answer

(i) A non-principal branch range of $\sin^{-1} x$ is $\left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$

(ii) A non-principal branch range of $\cos^{-1} x$ is $[\pi, 2\pi]$

Question 10

Find the maximum value of $(\cos^{-1} x)^2$.

TYPE 5 Extrema of Inverse Trigonometric Functions

To Find: The maximum value of $(\cos^{-1} x)^2$

Concept / Formula Used: $\cos^{-1} x \quad [0, \pi]$

Solution:

For any $x \in [-1, 1]$, the range of the principal value branch of $\cos^{-1} x$ is:

$$0 \le \cos^{-1} x \le \pi$$

Since both bounds are non-negative, squaring all parts preserves the inequality:

$$0^2 \le (\cos^{-1} x)^2 \le \pi^2$$
$$0 \le (\cos^{-1} x)^2 \le \pi^2$$

The maximum value occurs at $x = -1$, where $\cos^{-1}(-1) = \pi$:

$$\text{Maximum value} = \pi^2$$

Final Answer

The maximum value of $(\cos^{-1} x)^2$ is π^2.

Question 11

If $\cot^{-1}\left(\frac{3}{4}\right) = x$, find the value of $\cos x$.

TYPE 6 Evaluating Trigonometric Ratios from Inverse Functions

Given: $\cot^{-1}\left(\frac{3}{4}\right) = x$

To Find: The value of $\cos x$

Concept / Formula Used: $\cot x = \dfrac{\text{Base}}{\text{Perpendicular}} \qquad \cos x = \dfrac{\text{Base}}{\text{Hypotenuse}}$

Solution:

Given:

$$\cot^{-1}\left(\frac{3}{4}\right) = x \implies \cot x = \frac{3}{4}$$

Since $\frac{3}{4} > 0$, x lies in the first quadrant, i.e., $x \in \left(0, \frac{\pi}{2}\right)$.
Using the trigonometric identity $\operatorname{cosec}^2 x = 1 + \cot^2 x$:

$$\begin{aligned}\operatorname{cosec}^2 x &= 1 + \left(\frac{3}{4}\right)^2 \\ &= 1 + \frac{9}{16} \\ &= \frac{25}{16}\end{aligned}$$
$$\implies \sin^2 x = \frac{16}{25} \implies \sin x = \frac{4}{5} \quad (\text{as } x \in (0, \pi/2))$$

Now, evaluating $\cos x$:

$$\begin{aligned}\cos x &= \cot x \cdot \sin x \\ &= \frac{3}{4} \cdot \frac{4}{5} \\ &= \frac{3}{5}\end{aligned}$$

Final Answer

$\cos x = \frac{3}{5}$

Question 12

(i) If $\tan^{-1}\left(\frac{3}{4}\right) = x$, find the values of $\cos x$ and $\sin x$.

(ii) If $\cot^{-1}\left(-\frac{1}{7}\right) = x$, find the values of $\sin x$ and $\cos x$.

(iii) If $\tan^{-1} x = \sin^{-1}\left(\frac{1}{2}\right)$, find the value of x.

TYPE 6 Evaluating Trigonometric Ratios from Inverse Functions

To Find: The required trigonometric and algebraic values

Solution:

(i) Given $\tan^{-1}\left(\frac{3}{4}\right) = x \implies \tan x = \frac{3}{4}$, with $x \in \left(0, \frac{\pi}{2}\right)$.
Using $\sec^2 x = 1 + \tan^2 x$:

$$\sec^2 x = 1 + \left(\frac{3}{4}\right)^2 = 1 + \frac{9}{16} = \frac{25}{16}$$
$$\sec x = \frac{5}{4} \implies \cos x = \frac{4}{5}$$

Then:

$$\sin x = \tan x \cdot \cos x = \frac{3}{4} \cdot \frac{4}{5} = \frac{3}{5}$$

(ii) Given $\cot^{-1}\left(-\frac{1}{7}\right) = x \implies \cot x = -\frac{1}{7}$.

Since the principal branch of $\cot^{-1}$ is $(0, \pi)$ and the argument is negative, $x \in \left(\frac{\pi}{2}, \pi\right)$ (Quadrant II).

Using $\operatorname{cosec}^2 x = 1 + \cot^2 x$:

$$\operatorname{cosec}^2 x = 1 + \left(-\frac{1}{7}\right)^2 = 1 + \frac{1}{49} = \frac{50}{49}$$

$$\operatorname{cosec} x = \frac{\sqrt{50}}{7} = \frac{5\sqrt{2}}{7} \quad \text{(since } \sin x > 0 \text{ in Quadrant II)}$$

$$\implies \sin x = \frac{7}{5\sqrt{2}} = \frac{7\sqrt{2}}{10}$$

Now, $\cos x = \cot x \cdot \sin x$:

$$\cos x = \left(-\frac{1}{7}\right) \cdot \left(\frac{7}{5\sqrt{2}}\right) = -\frac{1}{5\sqrt{2}} = -\frac{\sqrt{2}}{10}$$

(iii) Given $\tan^{-1} x = \sin^{-1}\left(\frac{1}{2}\right)$.

First, evaluate the RHS:

$$\sin^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{6}$$

Therefore:

$$\tan^{-1} x = \frac{\pi}{6}$$

$$x = \tan\left(\frac{\pi}{6}\right)$$

$$x = \frac{1}{\sqrt{3}}$$

Final Answer

(i) $\cos x = \frac{4}{5}$, $\sin x = \frac{3}{5}$

(ii) $\sin x = \frac{7}{5\sqrt{2}} = \frac{7\sqrt{2}}{10}$, $\cos x = -\frac{1}{5\sqrt{2}} = -\frac{\sqrt{2}}{10}$

(iii) $x = \frac{1}{\sqrt{3}}$

Question 13

Evaluate the following:

(i) $\cot(\tan^{-1}\sqrt{3})$

(ii) $\sin\left(\frac{\pi}{6} - \sin^{-1}\left(-\frac{\sqrt{3}}{2}\right)\right)$

(iii) $\cos\left(\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right) + \frac{\pi}{6}\right)$

TYPE 7 **Evaluating Expressions with Inverse Trigonometric Functions**

To Find: The numerical value of each expression

Concept / Formula Used: $\cot(\tan^{-1} u) = \frac{1}{u}$ $\sin^{-1}(-u) = -\sin^{-1} u$ $\cos^{-1}(-u) = \pi - \cos^{-1} u$

Solution:

(i) Since $\tan^{-1}\sqrt{3} = \frac{\pi}{3}$:

$$\begin{aligned}\cot(\tan^{-1}\sqrt{3}) &= \cot\left(\frac{\pi}{3}\right)\\ &= \frac{1}{\sqrt{3}}\end{aligned}$$

(ii) First, evaluate $\sin^{-1}\left(-\frac{\sqrt{3}}{2}\right)$:

$$\sin^{-1}\left(-\frac{\sqrt{3}}{2}\right) = -\sin^{-1}\left(\frac{\sqrt{3}}{2}\right) = -\frac{\pi}{3}$$

Substituting this into the expression:

$$\begin{aligned}\sin\left(\frac{\pi}{6} - \sin^{-1}\left(-\frac{\sqrt{3}}{2}\right)\right) &= \sin\left(\frac{\pi}{6} - \left(-\frac{\pi}{3}\right)\right)\\ &= \sin\left(\frac{\pi}{6} + \frac{\pi}{3}\right)\\ &= \sin\left(\frac{3\pi}{6}\right)\\ &= \sin\left(\frac{\pi}{2}\right)\\ &= 1\end{aligned}$$

(iii) First, evaluate $\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right)$:

$$\begin{aligned}\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right) &= \pi - \cos^{-1}\left(\frac{\sqrt{3}}{2}\right)\\ &= \pi - \frac{\pi}{6}\\ &= \frac{5\pi}{6}\end{aligned}$$

Substituting this into the expression:

$$\begin{aligned}\cos\left(\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right)+\frac{\pi}{6}\right) &= \cos\left(\frac{5\pi}{6}+\frac{\pi}{6}\right)\\ &= \cos\left(\frac{6\pi}{6}\right)\\ &= \cos(\pi)\\ &= -1\end{aligned}$$

Final Answer

(i) $\cot(\tan^{-1}\sqrt{3}) = \frac{1}{\sqrt{3}}$

(ii) $\sin\left(\frac{\pi}{6}-\sin^{-1}\left(-\frac{\sqrt{3}}{2}\right)\right) = 1$

(iii) $\cos\left(\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right)+\frac{\pi}{6}\right) = -1$

Question 14

Prove the following:

(i) $\sin^{-1}\left(-\frac{1}{2}\right)+\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right) = \frac{2\pi}{3}$

(ii) $\tan^{-1}(-1)+\cos^{-1}\left(-\frac{1}{\sqrt{2}}\right) = \frac{\pi}{2}$

TYPE 7 **Evaluating Expressions with Inverse Trigonometric Functions**

To Prove: The given inverse trigonometric equations hold true

Concept / Formula Used: $\sin^{-1}(-x) = -\sin^{-1}x$ $\cos^{-1}(-x) = \pi - \cos^{-1}x$ $\tan^{-1}(-x) = -\tan^{-1}x$

Solution:

(i) Evaluating each term on the LHS:

$$\sin^{-1}\left(-\frac{1}{2}\right) = -\sin^{-1}\left(\frac{1}{2}\right) = -\frac{\pi}{6}$$
$$\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right) = \pi-\cos^{-1}\left(\frac{\sqrt{3}}{2}\right) = \pi-\frac{\pi}{6} = \frac{5\pi}{6}$$

Adding the two terms:

$$\begin{aligned}\text{LHS} &= -\frac{\pi}{6} + \frac{5\pi}{6} \\ &= \frac{4\pi}{6} \\ &= \frac{2\pi}{3} \\ &= \text{RHS}\end{aligned}$$

(ii) Evaluating each term on the LHS:

$$\tan^{-1}(-1) = -\tan^{-1}(1) = -\frac{\pi}{4}$$

$$\cos^{-1}\left(-\frac{1}{\sqrt{2}}\right) = \pi - \cos^{-1}\left(\frac{1}{\sqrt{2}}\right) = \pi - \frac{\pi}{4} = \frac{3\pi}{4}$$

Adding the two terms:

$$\begin{aligned}\text{LHS} &= -\frac{\pi}{4} + \frac{3\pi}{4} \\ &= \frac{2\pi}{4} \\ &= \frac{\pi}{2} \\ &= \text{RHS}\end{aligned}$$

Hence Proved

In both cases, LHS = RHS, hence proved.

Question 15

Using principal values, find the values of:

(i) $\cos^{-1}\left(\frac{1}{2}\right) - 2\sin^{-1}\left(-\frac{1}{2}\right)$

(ii) $\tan^{-1}(\sqrt{3}) - \cot^{-1}(-\sqrt{3})$

(iii) $\tan^{-1}(1) + \cos^{-1}\left(-\frac{1}{2}\right)$

(iv) $\operatorname{cosec}^{-1}(-1) + \cot^{-1}\left(-\frac{1}{\sqrt{3}}\right)$

TYPE 7 **Evaluating Expressions with Inverse Trigonometric Functions**

To Find: The values of the given expressions

Solution:

(i) Evaluating individual terms:

$$\cos^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{3}$$

$$\sin^{-1}\left(-\frac{1}{2}\right) = -\frac{\pi}{6}$$

Substituting these into the expression:

$$\begin{aligned}\cos^{-1}\left(\frac{1}{2}\right) - 2\sin^{-1}\left(-\frac{1}{2}\right) &= \frac{\pi}{3} - 2\left(-\frac{\pi}{6}\right)\\ &= \frac{\pi}{3} + \frac{\pi}{3}\\ &= \frac{2\pi}{3}\end{aligned}$$

(ii) Evaluating individual terms:

$$\begin{aligned}\tan^{-1}(\sqrt{3}) &= \frac{\pi}{3}\\ \cot^{-1}(-\sqrt{3}) &= \pi - \cot^{-1}(\sqrt{3}) = \pi - \frac{\pi}{6} = \frac{5\pi}{6}\end{aligned}$$

Substituting these into the expression:

$$\begin{aligned}\tan^{-1}(\sqrt{3}) - \cot^{-1}(-\sqrt{3}) &= \frac{\pi}{3} - \frac{5\pi}{6}\\ &= \frac{2\pi - 5\pi}{6}\\ &= -\frac{3\pi}{6}\\ &= -\frac{\pi}{2}\end{aligned}$$

(iii) Evaluating individual terms:

$$\begin{aligned}\tan^{-1}(1) &= \frac{\pi}{4}\\ \cos^{-1}\left(-\frac{1}{2}\right) &= \pi - \cos^{-1}\left(\frac{1}{2}\right) = \pi - \frac{\pi}{3} = \frac{2\pi}{3}\end{aligned}$$

Substituting these into the expression:

$$\begin{aligned}\tan^{-1}(1) + \cos^{-1}\left(-\frac{1}{2}\right) &= \frac{\pi}{4} + \frac{2\pi}{3}\\ &= \frac{3\pi + 8\pi}{12}\\ &= \frac{11\pi}{12}\end{aligned}$$

(iv) Evaluating individual terms:

$$\begin{aligned}\operatorname{cosec}^{-1}(-1) &= -\operatorname{cosec}^{-1}(1) = -\frac{\pi}{2}\\ \cot^{-1}\left(-\frac{1}{\sqrt{3}}\right) &= \pi - \cot^{-1}\left(\frac{1}{\sqrt{3}}\right) = \pi - \frac{\pi}{3} = \frac{2\pi}{3}\end{aligned}$$

Substituting these into the expression:

$$\begin{aligned}\operatorname{cosec}^{-1}(-1) + \cot^{-1}\left(-\frac{1}{\sqrt{3}}\right) &= -\frac{\pi}{2} + \frac{2\pi}{3} \\ &= \frac{-3\pi + 4\pi}{6} \\ &= \frac{\pi}{6}\end{aligned}$$

Final Answer

(i) $\dfrac{2\pi}{3}$

(ii) $-\dfrac{\pi}{2}$

(iii) $\dfrac{11\pi}{12}$

(iv) $\dfrac{\pi}{6}$

Question 16

Find the values of the following:

(i) $\tan^{-1}\left(2\cos\left(2\sin^{-1}\frac{1}{2}\right)\right)$

(ii) $\tan^{-1}\left(2\sin\left(2\cos^{-1}\frac{\sqrt{3}}{2}\right)\right)$

(iii) $\sin^{-1}\left(\frac{1}{2}\sin\left(2\cot^{-1}(-1)\right)\right)$

(iv) $\cot(\sin^{-1}(\cos(\tan^{-1}1)))$

TYPE 8 **Evaluating Nested Inverse Trigonometric Functions**

★ **Likely Exam Question**

To Find: The values of the nested trigonometric expressions

Solution:

(i) Innermost term: $\sin^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{6}$.

Working outwards:

$$\begin{aligned}2\sin^{-1}\left(\frac{1}{2}\right) &= 2\cdot\frac{\pi}{6} = \frac{\pi}{3} \\ \cos\left(\frac{\pi}{3}\right) &= \frac{1}{2} \\ 2\cos\left(\frac{\pi}{3}\right) &= 2\cdot\frac{1}{2} = 1 \\ \tan^{-1}(1) &= \frac{\pi}{4}\end{aligned}$$

(ii) Innermost term: $\cos^{-1}\left(\frac{\sqrt{3}}{2}\right) = \frac{\pi}{6}$.

Working outwards:

$$2\cos^{-1}\left(\frac{\sqrt{3}}{2}\right) = 2 \cdot \frac{\pi}{6} = \frac{\pi}{3}$$
$$\sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}$$
$$2\sin\left(\frac{\pi}{3}\right) = 2 \cdot \frac{\sqrt{3}}{2} = \sqrt{3}$$
$$\tan^{-1}(\sqrt{3}) = \frac{\pi}{3}$$

(iii) Innermost term: $\cot^{-1}(-1) = \pi - \cot^{-1}(1) = \pi - \frac{\pi}{4} = \frac{3\pi}{4}$.
Working outwards:

$$2\cot^{-1}(-1) = 2 \cdot \frac{3\pi}{4} = \frac{3\pi}{2}$$
$$\sin\left(\frac{3\pi}{2}\right) = -1$$
$$\frac{1}{2}\sin\left(\frac{3\pi}{2}\right) = \frac{1}{2}(-1) = -\frac{1}{2}$$
$$\sin^{-1}\left(-\frac{1}{2}\right) = -\frac{\pi}{6}$$

(iv) Innermost term: $\tan^{-1}(1) = \frac{\pi}{4}$.
Working outwards:

$$\cos(\tan^{-1} 1) = \cos\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$$
$$\sin^{-1}\left(\frac{1}{\sqrt{2}}\right) = \frac{\pi}{4}$$
$$\cot\left(\frac{\pi}{4}\right) = 1$$

Final Answer

(i) $\frac{\pi}{4}$

(ii) $\frac{\pi}{3}$

(iii) $-\frac{\pi}{6}$

(iv) 1

Question 17

Find the value of:

(i) $\tan^{-1}\left(\tan\frac{5\pi}{6}\right)+\cos^{-1}\left(\cos\frac{13\pi}{6}\right)$

(ii) $\tan^{-1}\left(-\frac{1}{\sqrt{3}}\right)+\cot^{-1}\left(\frac{1}{\sqrt{3}}\right)+\tan^{-1}\left(\sin\left(-\frac{\pi}{2}\right)\right)$

TYPE 2 **Evaluating Composite Inverse Trigonometric Functions**

★ **Likely Exam Question**

To Find: The values of the given expressions

Solution:

(i) First evaluate each term individually:

$$\tan\left(\frac{5\pi}{6}\right)=\tan\left(\pi-\frac{\pi}{6}\right)=-\tan\left(\frac{\pi}{6}\right)=\tan\left(-\frac{\pi}{6}\right)$$

$$\implies \tan^{-1}\left(\tan\frac{5\pi}{6}\right)=\tan^{-1}\left(\tan\left(-\frac{\pi}{6}\right)\right)=-\frac{\pi}{6}\quad\left(\text{since }-\frac{\pi}{6}\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right)\right)$$

For the second term:

$$\cos\left(\frac{13\pi}{6}\right)=\cos\left(2\pi+\frac{\pi}{6}\right)=\cos\left(\frac{\pi}{6}\right)$$

$$\implies \cos^{-1}\left(\cos\frac{13\pi}{6}\right)=\cos^{-1}\left(\cos\frac{\pi}{6}\right)=\frac{\pi}{6}\quad\left(\text{since }\frac{\pi}{6}\in[0,\pi]\right)$$

Summing the terms:

$$\tan^{-1}\left(\tan\frac{5\pi}{6}\right)+\cos^{-1}\left(\cos\frac{13\pi}{6}\right)=-\frac{\pi}{6}+\frac{\pi}{6}=0$$

(ii) Evaluate each term individually:

$$\tan^{-1}\left(-\frac{1}{\sqrt{3}}\right)=-\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)=-\frac{\pi}{6}$$

$$\cot^{-1}\left(\frac{1}{\sqrt{3}}\right)=\frac{\pi}{3}$$

$$\sin\left(-\frac{\pi}{2}\right)=-1\implies\tan^{-1}\left(\sin\left(-\frac{\pi}{2}\right)\right)=\tan^{-1}(-1)=-\frac{\pi}{4}$$

Summing the three terms:

$$\begin{aligned}\text{Value}&=-\frac{\pi}{6}+\frac{\pi}{3}-\frac{\pi}{4}\\&=\frac{-2\pi+4\pi-3\pi}{12}\\&=\frac{-\pi}{12}\\&=-\frac{\pi}{12}\end{aligned}$$

Final Answer

(i) 0

(ii) $-\dfrac{\pi}{12}$

Question 18 (i)

Evaluate:

$$\sin^{-1}\left(\sin\frac{3\pi}{4}\right) + \cos^{-1}(\cos\pi) + \tan^{-1}(1)$$

TYPE 7 Evaluating Expressions with Inverse Trigonometric Functions

To Find: The value of $\sin^{-1}\left(\sin\frac{3\pi}{4}\right) + \cos^{-1}(\cos\pi) + \tan^{-1}(1)$

Concept / Formula Used: $\sin^{-1}(\sin x) = x \qquad x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \qquad \cos^{-1}(\cos x) = x \qquad x \in [0, \pi]$

Solution:

We evaluate each term separately using the principal value branches of inverse trigonometric functions.

First term:

$$\text{Since } \frac{3\pi}{4} \notin \left[-\frac{\pi}{2}, \frac{\pi}{2}\right], \text{ we rewrite } \sin\frac{3\pi}{4}:$$
$$\sin\frac{3\pi}{4} = \sin\left(\pi - \frac{\pi}{4}\right) = \sin\frac{\pi}{4}$$
$$\implies \sin^{-1}\left(\sin\frac{3\pi}{4}\right) = \sin^{-1}\left(\sin\frac{\pi}{4}\right) = \frac{\pi}{4} \quad \left(\text{since } \frac{\pi}{4} \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]\right)$$

Second term:

$$\text{Since } \pi \in [0, \pi], \text{ we have:}$$
$$\cos^{-1}(\cos\pi) = \pi$$

Third term:

$$\text{Since } \tan\frac{\pi}{4} = 1 \text{ and } \frac{\pi}{4} \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right):$$
$$\tan^{-1}(1) = \frac{\pi}{4}$$

Substituting these values back into the expression:

$$\begin{aligned}\sin^{-1}\left(\sin\frac{3\pi}{4}\right) + \cos^{-1}(\cos\pi) + \tan^{-1}(1) &= \frac{\pi}{4} + \pi + \frac{\pi}{4} \\ &= \frac{\pi}{2} + \pi \\ &= \frac{3\pi}{2}\end{aligned}$$

Final Answer

$\sin^{-1}\left(\sin\frac{3\pi}{4}\right) + \cos^{-1}(\cos\pi) + \tan^{-1}(1) = \frac{3\pi}{2}$

Common Mistake: A common error is writing $\sin^{-1}\left(\sin\frac{3\pi}{4}\right) = \frac{3\pi}{4}$ directly, ignoring the principal value range $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ for $\sin^{-1}$.

Question 18 (ii)

Evaluate:

$$\sin^{-1}\left(\sin\frac{3\pi}{4}\right) + \cos^{-1}\left(\cos\frac{3\pi}{4}\right) + \tan^{-1}(1)$$

TYPE 7 **Evaluating Expressions with Inverse Trigonometric Functions**

To Find: The value of $\sin^{-1}\left(\sin\frac{3\pi}{4}\right) + \cos^{-1}\left(\cos\frac{3\pi}{4}\right) + \tan^{-1}(1)$

Concept / Formula Used: $\sin^{-1}(\sin x) = x \quad x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \quad \cos^{-1}(\cos x) = x \quad x \in [0, \pi]$

Solution:

We evaluate each term using its respective principal value branch.

First term:

$$\sin\frac{3\pi}{4} = \sin\left(\pi - \frac{\pi}{4}\right) = \sin\frac{\pi}{4}$$

$$\implies \sin^{-1}\left(\sin\frac{3\pi}{4}\right) = \sin^{-1}\left(\sin\frac{\pi}{4}\right) = \frac{\pi}{4} \quad \left(\text{since } \frac{\pi}{4} \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]\right)$$

Second term:

$$\text{Since } \frac{3\pi}{4} \in [0, \pi], \text{ we have:}$$

$$\cos^{-1}\left(\cos\frac{3\pi}{4}\right) = \frac{3\pi}{4}$$

Third term:

$$\tan^{-1}(1) = \frac{\pi}{4} \quad \left(\text{since } \tan\frac{\pi}{4} = 1 \text{ and } \frac{\pi}{4} \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)\right)$$

Adding all three terms together:

$$\begin{aligned}\sin^{-1}\left(\sin\frac{3\pi}{4}\right) + \cos^{-1}\left(\cos\frac{3\pi}{4}\right) + \tan^{-1}(1) &= \frac{\pi}{4} + \frac{3\pi}{4} + \frac{\pi}{4} \\ &= \pi + \frac{\pi}{4} \\ &= \frac{5\pi}{4}\end{aligned}$$

Final Answer

$$\sin^{-1}\left(\sin\frac{3\pi}{4}\right) + \cos^{-1}\left(\cos\frac{3\pi}{4}\right) + \tan^{-1}(1) = \frac{5\pi}{4}$$

Question 18 (iii)

Evaluate:

$$3\sin^{-1}\left(\frac{1}{\sqrt{2}}\right) + 2\cos^{-1}\left(\frac{\sqrt{3}}{2}\right) + \cos^{-1}(0)$$

TYPE 7 Evaluating Expressions with Inverse Trigonometric Functions

★ Likely Exam Question

To Find: The value of $3\sin^{-1}\left(\frac{1}{\sqrt{2}}\right) + 2\cos^{-1}\left(\frac{\sqrt{3}}{2}\right) + \cos^{-1}(0)$

Concept / Formula Used: $\sin^{-1}(y) = \theta \iff \sin\theta = y \quad \theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ $\quad \cos^{-1}(y) = \phi \iff \cos\phi = y \quad \phi \in [0, \pi]$

Solution:

Evaluating each term using principal values:

First term:

$$\text{Let } \sin^{-1}\left(\frac{1}{\sqrt{2}}\right) = \theta$$
$$\implies \sin\theta = \frac{1}{\sqrt{2}} \implies \theta = \frac{\pi}{4}$$
$$\implies 3\sin^{-1}\left(\frac{1}{\sqrt{2}}\right) = 3 \times \frac{\pi}{4} = \frac{3\pi}{4}$$

Second term:

$$\text{Let } \cos^{-1}\left(\frac{\sqrt{3}}{2}\right) = \phi$$
$$\implies \cos\phi = \frac{\sqrt{3}}{2} \implies \phi = \frac{\pi}{6}$$
$$\implies 2\cos^{-1}\left(\frac{\sqrt{3}}{2}\right) = 2 \times \frac{\pi}{6} = \frac{\pi}{3}$$

Third term:

$$\text{Let } \cos^{-1}(0) = \alpha$$
$$\implies \cos\alpha = 0 \implies \alpha = \frac{\pi}{2}$$

Combining all three parts:

$$3\sin^{-1}\left(\frac{1}{\sqrt{2}}\right) + 2\cos^{-1}\left(\frac{\sqrt{3}}{2}\right) + \cos^{-1}(0) = \frac{3\pi}{4} + \frac{\pi}{3} + \frac{\pi}{2}$$
$$= \frac{9\pi + 4\pi + 6\pi}{12}$$
$$= \frac{19\pi}{12}$$

Final Answer

$$3\sin^{-1}\left(\frac{1}{\sqrt{2}}\right) + 2\cos^{-1}\left(\frac{\sqrt{3}}{2}\right) + \cos^{-1}(0) = \frac{19\pi}{12}$$

Question 19 (i)

Evaluate:

$$\cos\left(\sin^{-1}\left(-\frac{3}{5}\right)\right)$$

TYPE 9 **Evaluating Trigonometric Functions of Inverse Arguments**

To Find: The value of $\cos\left(\sin^{-1}\left(-\frac{3}{5}\right)\right)$

Concept / Formula Used: $\cos\theta = \sqrt{1-\sin^2\theta} \qquad \theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

Solution:

Let $\theta = \sin^{-1}\left(-\frac{3}{5}\right)$.

By the definition of inverse sine, we have:

$$\sin\theta = -\frac{3}{5} \quad \text{where } \theta \in \left[-\frac{\pi}{2}, 0\right)$$

Since θ lies in Quadrant IV, $\cos\theta > 0$. Using the fundamental trigonometric identity:

$$\cos\theta = \sqrt{1-\sin^2\theta}$$
$$= \sqrt{1-\left(-\frac{3}{5}\right)^2}$$
$$= \sqrt{1-\frac{9}{25}}$$
$$= \sqrt{\frac{25-9}{25}}$$
$$= \sqrt{\frac{16}{25}}$$
$$= \frac{4}{5}$$

Back-substituting $\theta = \sin^{-1}\left(-\frac{3}{5}\right)$:

$$\cos\left(\sin^{-1}\left(-\frac{3}{5}\right)\right) = \frac{4}{5}$$

Final Answer

$\cos\left(\sin^{-1}\left(-\frac{3}{5}\right)\right) = \frac{4}{5}$

Question 19 (ii)

Evaluate:

$$\operatorname{cosec}\left(\cos^{-1}\left(-\frac{12}{13}\right)\right)$$

TYPE 9 Evaluating Trigonometric Functions of Inverse Arguments

To Find: The value of $\operatorname{cosec}\left(\cos^{-1}\left(-\frac{12}{13}\right)\right)$

Concept / Formula Used: $\sin\theta = \sqrt{1-\cos^2\theta} \qquad \theta \in [0, \pi] \qquad \operatorname{cosec}\theta = \frac{1}{\sin\theta}$

Solution:

Let $\theta = \cos^{-1}\left(-\frac{12}{13}\right)$.

By definition of inverse cosine:

$$\cos\theta = -\frac{12}{13} \quad \text{where } \theta \in \left(\frac{\pi}{2}, \pi\right]$$

Since θ lies in Quadrant II, $\sin\theta$ is positive:

$$\begin{aligned}
\sin\theta &= \sqrt{1-\cos^2\theta} \\
&= \sqrt{1-\left(-\frac{12}{13}\right)^2} \\
&= \sqrt{1-\frac{144}{169}} \\
&= \sqrt{\frac{169-144}{169}} \\
&= \sqrt{\frac{25}{169}} \\
&= \frac{5}{13}
\end{aligned}$$

Now, using the reciprocal relation $\operatorname{cosec}\theta = \dfrac{1}{\sin\theta}$:

$$\begin{aligned}\operatorname{cosec}\theta &= \frac{1}{\frac{5}{13}} \\ &= \frac{13}{5}\end{aligned}$$

Back-substituting $\theta = \cos^{-1}\left(-\frac{12}{13}\right)$:

$$\operatorname{cosec}\left(\cos^{-1}\left(-\frac{12}{13}\right)\right) = \frac{13}{5}$$

Final Answer

$\operatorname{cosec}\left(\cos^{-1}\left(-\dfrac{12}{13}\right)\right) = \dfrac{13}{5}$

Question 19 (iii)

Evaluate:

$$\sin\left(2\cot^{-1}\frac{1}{5}\right)$$

TYPE 9 **Evaluating Trigonometric Functions of Inverse Arguments**

★ Likely Exam Question

To Find: The value of $\sin\left(2\cot^{-1}\frac{1}{5}\right)$

Concept / Formula Used: $\sin(2\theta) = \frac{2\tan\theta}{1+\tan^2\theta}$ $\tan(\cot^{-1}x) = \frac{1}{x}$

Solution:

Let $\theta = \cot^{-1}\frac{1}{5}$.

Then:

$$\begin{aligned}\cot\theta &= \frac{1}{5} \\ \implies \tan\theta &= \frac{1}{\cot\theta} = 5\end{aligned}$$

We need to compute $\sin(2\theta)$. Using the double-angle formula for sine in terms of tangent:

$$\sin(2\theta) = \frac{2\tan\theta}{1+\tan^2\theta}$$

Substituting $\tan\theta = 5$:

$$\begin{aligned}\sin(2\theta) &= \frac{2(5)}{1+(5)^2} \\ &= \frac{10}{1+25} \\ &= \frac{10}{26} \\ &= \frac{5}{13}\end{aligned}$$

Back-substituting $\theta = \cot^{-1}\frac{1}{5}$:

$$\sin\left(2\cot^{-1}\frac{1}{5}\right) = \frac{5}{13}$$

Final Answer

$\sin\left(2\cot^{-1}\frac{1}{5}\right) = \frac{5}{13}$

Question 20

Show that:

$$\tan\left(\frac{1}{2}\sin^{-1}\frac{3}{4}\right) = \frac{4-\sqrt{7}}{3}$$

TYPE 10 Half-Angle Substitution in Inverse Trigonometric Functions

★ **Likely Exam Question**

To Prove: $\tan\left(\frac{1}{2}\sin^{-1}\frac{3}{4}\right) = \frac{4-\sqrt{7}}{3}$

Concept / Formula Used: $\sin\theta = \frac{2t}{1+t^2}$ $\quad t = \tan\left(\frac{\theta}{2}\right)$

Solution:

Let $\theta = \sin^{-1}\frac{3}{4}$.

By definition of inverse sine:

$$\sin\theta = \frac{3}{4} \quad \text{where } \theta \in \left(0, \frac{\pi}{2}\right)$$

Let $t = \tan\left(\frac{\theta}{2}\right)$. Since $0 < \theta < \frac{\pi}{2}$, we have $0 < \frac{\theta}{2} < \frac{\pi}{4}$, which implies $0 < t < 1$.

Expressing $\sin\theta$ in terms of $t = \tan\left(\frac{\theta}{2}\right)$:

$$\begin{aligned}\sin\theta &= \frac{2t}{1+t^2} \\ \implies \frac{3}{4} &= \frac{2t}{1+t^2}\end{aligned}$$

Cross-multiplying to form a quadratic equation in t:

$$\begin{aligned}3(1+t^2) &= 4(2t)\\ 3+3t^2 &= 8t\\ 3t^2-8t+3 &= 0\end{aligned}$$

Solving for t using the quadratic formula $t = \frac{-b\pm\sqrt{b^2-4ac}}{2a}$ with $a = 3$, $b = -8$, $c = 3$:

$$\begin{aligned}t &= \frac{-(-8)\pm\sqrt{(-8)^2-4(3)(3)}}{2(3)}\\ &= \frac{8\pm\sqrt{64-36}}{6}\\ &= \frac{8\pm\sqrt{28}}{6}\\ &= \frac{8\pm 2\sqrt{7}}{6}\\ &= \frac{4\pm\sqrt{7}}{3}\end{aligned}$$

Since $0 < t < 1$ and $\sqrt{7} \approx 2.646$:

$$\begin{aligned}\frac{4+\sqrt{7}}{3} &\approx \frac{4+2.646}{3} = \frac{6.646}{3} \approx 2.215 > 1 \quad \text{(rejected)}\\ \frac{4-\sqrt{7}}{3} &\approx \frac{4-2.646}{3} = \frac{1.354}{3} \approx 0.451 \in (0,1) \quad \text{(accepted)}\end{aligned}$$

Alternatively, using the half-angle formula $\tan\left(\frac{\theta}{2}\right) = \frac{1-\cos\theta}{\sin\theta}$:

$$\begin{aligned}\cos\theta &= \sqrt{1-\sin^2\theta} = \sqrt{1-\frac{9}{16}} = \frac{\sqrt{7}}{4}\\ \implies \tan\left(\frac{\theta}{2}\right) &= \frac{1-\frac{\sqrt{7}}{4}}{\frac{3}{4}} = \frac{4-\sqrt{7}}{3}\end{aligned}$$

Back-substituting $\theta = \sin^{-1}\frac{3}{4}$:

$$\tan\left(\frac{1}{2}\sin^{-1}\frac{3}{4}\right) = \frac{4-\sqrt{7}}{3}$$

Hence Proved

LHS = RHS = $\frac{4-\sqrt{7}}{3}$, hence proved.

Common Mistake: Taking the positive root $\frac{4+\sqrt{7}}{3}$ without checking the domain condition $0 < \frac{\theta}{2} < \frac{\pi}{4} \implies 0 < t < 1$ is a common pitfall.

2.2 – Properties and Simplification of Inverse Trigonometric Functions

Question 1

Evaluate the following:

(i) $\sin^{-1}(\sin 2)$
(ii) $\sin^{-1}(\sin 5)$
(iii) $\tan^{-1}(\tan 4)$
(iv) $\cos^{-1}(\cos 10)$

TYPE 2 **Evaluating Composite Inverse Trigonometric Functions**

To Find: The principal value of each given inverse trigonometric expression.

Concept / Formula Used: $\sin^{-1}(\sin\theta) = \theta \iff \theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ $\quad \tan^{-1}(\tan\theta) = \theta \iff \theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ $\quad \cos^{-1}(\cos\theta) = \theta \iff \theta \in [0, \pi]$

Solution:

(i) For $\sin^{-1}(\sin 2)$:

$$\begin{aligned}
\text{Since } \frac{\pi}{2} \approx 1.57 \text{ and } \pi \approx 3.14, \text{ we have } 2 \in \left(\frac{\pi}{2}, \pi\right). \\
\text{Using the identity } \sin(\pi - \theta) = \sin\theta: \\
\sin(2) &= \sin(\pi - 2) \\
\text{Since } -\frac{\pi}{2} \le \pi - 2 \le \frac{\pi}{2}: \\
\sin^{-1}(\sin 2) &= \sin^{-1}(\sin(\pi - 2)) \\
&= \pi - 2
\end{aligned}$$

(ii) For $\sin^{-1}(\sin 5)$:

$$\begin{aligned}
\text{Since } \frac{3\pi}{2} \approx 4.71 \text{ and } 2\pi \approx 6.28, \text{ we have } 5 \in \left(\frac{3\pi}{2}, 2\pi\right). \\
\text{Using the identity } \sin(\theta - 2\pi) = \sin\theta: \\
\sin(5) &= \sin(5 - 2\pi) \\
\text{Since } -\frac{\pi}{2} \le 5 - 2\pi \le \frac{\pi}{2}: \\
\sin^{-1}(\sin 5) &= \sin^{-1}(\sin(5 - 2\pi)) \\
&= 5 - 2\pi
\end{aligned}$$

(iii) For $\tan^{-1}(\tan 4)$:

$$\text{Since } \pi \approx 3.14 \text{ and } \frac{3\pi}{2} \approx 4.71, \text{ we have } 4 \in \left(\pi, \frac{3\pi}{2}\right).$$

$$\begin{aligned} \text{Using the periodic identity } \tan(\theta - \pi) = \tan\theta : \\ \tan(4) &= \tan(4 - \pi) \\ \text{Since } -\frac{\pi}{2} < 4 - \pi < \frac{\pi}{2} : \\ \tan^{-1}(\tan 4) &= \tan^{-1}(\tan(4 - \pi)) \\ &= 4 - \pi \end{aligned}$$

(iv) For $\cos^{-1}(\cos 10)$:

$$\text{Since } 3\pi \approx 9.42 \text{ and } 4\pi \approx 12.57, \text{ we have } 10 \in (3\pi, 4\pi).$$

$$\begin{aligned} \text{Using the identity } \cos(4\pi - \theta) = \cos\theta : \\ \cos(10) &= \cos(4\pi - 10) \\ \text{Since } 0 \le 4\pi - 10 \le \pi : \\ \cos^{-1}(\cos 10) &= \cos^{-1}(\cos(4\pi - 10)) \\ &= 4\pi - 10 \end{aligned}$$

Final Answer

(i) $\pi - 2$ (ii) $5 - 2\pi$ (iii) $4 - \pi$ (iv) $4\pi - 10$

Common Mistake: Directly writing $\sin^{-1}(\sin x) = x$ without checking whether x lies in the principal value interval $[-\pi/2, \pi/2]$ leads to incorrect answers.

Question 2

Find the values of:

(i) $\sin(\sin^{-1} x + \cos^{-1} x)$, $|x| \le 1$

(ii) $\cot(\tan^{-1} x + \cot^{-1} x)$, $x \in \mathbb{R}$

TYPE 7 **Evaluating Expressions with Inverse Trigonometric Functions**

To Find: The value of each trigonometric expression.

Concept / Formula Used: $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$, for $|x| \le 1$ $\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$, for $x \in \mathbb{R}$

Solution:

(i) Using the standard result $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$:

$$\begin{aligned} \sin(\sin^{-1} x + \cos^{-1} x) &= \sin\left(\frac{\pi}{2}\right) \\ &= 1 \end{aligned}$$

(ii) Using the standard result $\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$:

$$\begin{aligned} \cot(\tan^{-1} x + \cot^{-1} x) &= \cot\left(\frac{\pi}{2}\right) \\ &= 0 \end{aligned}$$

Final Answer

(i) 1 (ii) 0

Question 3

Find the values of:

(i) $\csc(\sin^{-1} x + \cos^{-1} x), \quad |x| \leq 1$

(ii) $\cos(\sec^{-1} x + \csc^{-1} x), \quad |x| \geq 1$

TYPE 7 **Evaluating Expressions with Inverse Trigonometric Functions**

To Find: The value of each expression.

Concept / Formula Used: $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}, \quad |x| \leq 1 \qquad \sec^{-1} x + \csc^{-1} x = \frac{\pi}{2}, \quad |x| \geq 1$

Solution:

(i) Using $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$:

$$\begin{aligned} \csc(\sin^{-1} x + \cos^{-1} x) &= \csc\left(\frac{\pi}{2}\right) \\ &= 1 \end{aligned}$$

(ii) Using $\sec^{-1} x + \csc^{-1} x = \frac{\pi}{2}$:

$$\begin{aligned} \cos(\sec^{-1} x + \csc^{-1} x) &= \cos\left(\frac{\pi}{2}\right) \\ &= 0 \end{aligned}$$

Final Answer

(i) 1 (ii) 0

Question 4

Find the values of:

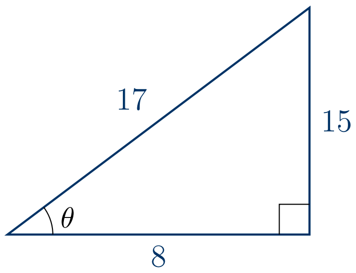
(i) $\tan^{-1}\left(\sin\left(-\dfrac{\pi}{2}\right)\right)$

(ii) $\cos\left(\tan^{-1}\dfrac{3}{4}\right)$

(iii) $\tan\left(\cos^{-1}\dfrac{8}{17}\right)$

TYPE 9 **Evaluating Trigonometric Functions of Inverse Arguments**

★ Likely Exam Question



Right triangle construction for $\theta = \cos^{-1}\left(\frac{8}{17}\right)$

To Find: The exact value of each trigonometric composite function.

Concept / Formula Used: $\theta = \cos^{-1}\left(\dfrac{a}{c}\right)$ Base $= a$ Hypotenuse $= c$ Perpendicular $= \sqrt{c^2 - a^2}$ $\tan\theta = \dfrac{\sqrt{c^2 - a^2}}{a}$

Solution:

(i) Evaluating inner function first:

$$\sin\left(-\frac{\pi}{2}\right) = -1$$

$$\tan^{-1}\left(\sin\left(-\frac{\pi}{2}\right)\right) = \tan^{-1}(-1) = -\frac{\pi}{4}$$

(ii) Let $\theta = \tan^{-1}\dfrac{3}{4} \implies \tan\theta = \dfrac{3}{4}$ with $\theta \in \left(0, \frac{\pi}{2}\right)$.

$$\text{Perpendicular} = 3, \quad \text{Base} = 4$$
$$\text{Hypotenuse} = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = 5$$
$$\cos\theta = \frac{\text{Base}}{\text{Hypotenuse}} = \frac{4}{5}$$
$$\therefore \cos\left(\tan^{-1}\frac{3}{4}\right) = \frac{4}{5}$$

(iii) Let $\phi = \cos^{-1}\dfrac{8}{17} \implies \cos\phi = \dfrac{8}{17}$ with $\phi \in \left(0, \frac{\pi}{2}\right)$.

$$\text{Base} = 8, \quad \text{Hypotenuse} = 17$$
$$\text{Perpendicular} = \sqrt{17^2 - 8^2} = \sqrt{289 - 64} = \sqrt{225} = 15$$
$$\tan\phi = \frac{\text{Perpendicular}}{\text{Base}} = \frac{15}{8}$$
$$\therefore \tan\left(\cos^{-1}\frac{8}{17}\right) = \frac{15}{8}$$

Final Answer

(i) $-\dfrac{\pi}{4}$ (ii) $\dfrac{4}{5}$ (iii) $\dfrac{15}{8}$

Question 5

Simplify the following:

(i) $\tan^{-1}\left(\dfrac{\sin x}{1 + \cos x}\right)$

(ii) $\tan^{-1}\left(\dfrac{x}{\sqrt{1 - x^2}}\right), \quad |x| < 1$

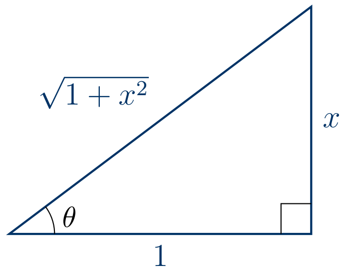
(iii) $\sin^{-1}\left(\dfrac{x}{\sqrt{1 + x^2}}\right)$

(iv) $\csc^{-1}\left(\dfrac{\sqrt{1 + x^2}}{x}\right)$

(v) $\cot^{-1}\left(\dfrac{x}{\sqrt{1 - x^2}}\right), \quad |x| < 1$

(vi) $\csc^{-1}\left(\dfrac{1}{\sqrt{1 - x^2}}\right), \quad |x| < 1$

TYPE 11 Simplifying Inverse Trigonometric Expressions



Right triangle for substitution $x = \tan\theta$ *in* $\sin^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right)$

To Find: The simplified form of each expression.

Concept / Formula Used: $\sin x = 2\sin\dfrac{x}{2}\cos\dfrac{x}{2}$ $1 + \cos x = 2\cos^2\dfrac{x}{2}$ $\sqrt{1-x^2}$ $x = \sin\theta$ $x = \cos\theta$ $\sqrt{1+x^2}$ $x = \tan\theta$

Solution:

(i) Using double/half-angle formulas:

$$\begin{aligned}\tan^{-1}\left(\frac{\sin x}{1+\cos x}\right) &= \tan^{-1}\left(\frac{2\sin\frac{x}{2}\cos\frac{x}{2}}{2\cos^2\frac{x}{2}}\right)\\ &= \tan^{-1}\left(\frac{\sin\frac{x}{2}}{\cos\frac{x}{2}}\right)\\ &= \tan^{-1}\left(\tan\frac{x}{2}\right)\\ &= \frac{x}{2}\end{aligned}$$

(ii) Let $x = \sin\theta \implies \theta = \sin^{-1}x$.

$$\begin{aligned}\sqrt{1-x^2} &= \sqrt{1-\sin^2\theta} = \cos\theta\\ \tan^{-1}\left(\frac{x}{\sqrt{1-x^2}}\right) &= \tan^{-1}\left(\frac{\sin\theta}{\cos\theta}\right)\\ &= \tan^{-1}(\tan\theta)\\ &= \theta\\ &= \sin^{-1}x\end{aligned}$$

(iii) Let $x = \tan\theta \implies \theta = \tan^{-1} x$.

$$\begin{aligned}
\sqrt{1+x^2} &= \sqrt{1+\tan^2\theta} = \sec\theta \\
\sin^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right) &= \sin^{-1}\left(\frac{\tan\theta}{\sec\theta}\right) \\
&= \sin^{-1}\left(\frac{\sin\theta/\cos\theta}{1/\cos\theta}\right) \\
&= \sin^{-1}(\sin\theta) \\
&= \theta \\
&= \tan^{-1} x
\end{aligned}$$

(iv) Using the identity $\csc^{-1}(y) = \sin^{-1}\left(\frac{1}{y}\right)$:

$$\begin{aligned}
\csc^{-1}\left(\frac{\sqrt{1+x^2}}{x}\right) &= \sin^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right) \\
&= \tan^{-1} x \quad \text{(from part iii)}
\end{aligned}$$

(v) Let $x = \cos\theta \implies \theta = \cos^{-1} x$.

$$\begin{aligned}
\sqrt{1-x^2} &= \sqrt{1-\cos^2\theta} = \sin\theta \\
\cot^{-1}\left(\frac{x}{\sqrt{1-x^2}}\right) &= \cot^{-1}\left(\frac{\cos\theta}{\sin\theta}\right) \\
&= \cot^{-1}(\cot\theta) \\
&= \theta \\
&= \cos^{-1} x
\end{aligned}$$

(vi) Using $\csc^{-1}(y) = \sin^{-1}\left(\frac{1}{y}\right)$:

$$\begin{aligned}
\csc^{-1}\left(\frac{1}{\sqrt{1-x^2}}\right) &= \sin^{-1}\left(\sqrt{1-x^2}\right) \\
\text{Let } x = \sin\theta &\implies \theta = \sin^{-1} x \\
\sin^{-1}\left(\sqrt{1-\sin^2\theta}\right) &= \sin^{-1}(\cos\theta) \\
&= \sin^{-1}\left(\sin\left(\frac{\pi}{2}-\theta\right)\right) \\
&= \frac{\pi}{2} - \theta \\
&= \frac{\pi}{2} - \sin^{-1} x \\
&= \cos^{-1} x
\end{aligned}$$

Final Answer

(i) $\dfrac{x}{2}$ (ii) $\sin^{-1} x$ (iii) $\tan^{-1} x$ (iv) $\tan^{-1} x$ (v) $\cos^{-1} x$ (vi) $\cos^{-1} x$

Question 6

Solve the following equations for x:

(i) $\tan^{-1} x = \sin^{-1}\left(\frac{1}{\sqrt{2}}\right)$

(ii) $\sin^{-1} x = \cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$

(iii) $\cos^{-1} x = \sin^{-1}\left(-\frac{1}{2}\right)$

(iv) $\cos(\sin^{-1} x) = \frac{1}{2}$

(v) $4\sin^{-1} x + \cos^{-1} x = \pi$

(vi) $\sin^{-1} x - \cos^{-1} x = \frac{\pi}{6}$

(vii) $5\tan^{-1} x + 3\cot^{-1} x = 2\pi$

(viii) $\tan^{-1} x^{-1} = \cot^{-1}\left(\frac{4}{x}\right), \quad x > 0$

TYPE 12 **Solving Inverse Trigonometric Equations**

★ **Likely Exam Question**

To Find: The value(s) of x satisfying each given equation.

Concept / Formula Used: $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$ $\quad \tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$ $\quad \tan^{-1}\left(\frac{1}{x}\right) = \cot^{-1} x \quad \text{for } x > 0$

Solution:

(i) Standard principal value:

$$\sin^{-1}\left(\frac{1}{\sqrt{2}}\right) = \frac{\pi}{4}$$
$$\tan^{-1} x = \frac{\pi}{4}$$
$$x = \tan\left(\frac{\pi}{4}\right)$$
$$x = 1$$

(ii) Standard principal value:

$$\begin{aligned}\cos^{-1}\left(\frac{\sqrt{3}}{2}\right) &= \frac{\pi}{6}\\ \sin^{-1}x &= \frac{\pi}{6}\\ x &= \sin\left(\frac{\pi}{6}\right)\\ x &= \frac{1}{2}\end{aligned}$$

(iii) Standard principal value:

$$\begin{aligned}\sin^{-1}\left(-\frac{1}{2}\right) &= -\frac{\pi}{6}\\ \cos^{-1}x &= -\frac{\pi}{6}\end{aligned}$$

Since the range of $\cos^{-1}x$ is $[0, \pi]$, $\cos^{-1}x$ cannot be negative. Therefore, there is no real solution for x.

(iv) Taking cosine inverse on both sides:

$$\begin{aligned}\cos(\sin^{-1}x) &= \frac{1}{2}\\ \sin^{-1}x &= \pm\frac{\pi}{3}\\ x &= \sin\left(\pm\frac{\pi}{3}\right)\\ x &= \pm\frac{\sqrt{3}}{2}\end{aligned}$$

(v) Using $\cos^{-1}x = \frac{\pi}{2} - \sin^{-1}x$:

$$\begin{aligned}4\sin^{-1}x + \left(\frac{\pi}{2} - \sin^{-1}x\right) &= \pi\\ 3\sin^{-1}x + \frac{\pi}{2} &= \pi\\ 3\sin^{-1}x &= \pi - \frac{\pi}{2}\\ 3\sin^{-1}x &= \frac{\pi}{2}\\ \sin^{-1}x &= \frac{\pi}{6}\\ x &= \sin\left(\frac{\pi}{6}\right)\\ x &= \frac{1}{2}\end{aligned}$$

(vi) Using $\cos^{-1} x = \frac{\pi}{2} - \sin^{-1} x$:

$$\begin{aligned}
\sin^{-1} x - \left(\frac{\pi}{2} - \sin^{-1} x\right) &= \frac{\pi}{6} \\
2\sin^{-1} x - \frac{\pi}{2} &= \frac{\pi}{6} \\
2\sin^{-1} x &= \frac{\pi}{6} + \frac{\pi}{2} \\
2\sin^{-1} x &= \frac{2\pi}{3} \\
\sin^{-1} x &= \frac{\pi}{3} \\
x &= \sin\left(\frac{\pi}{3}\right) \\
x &= \frac{\sqrt{3}}{2}
\end{aligned}$$

(vii) Using $\cot^{-1} x = \frac{\pi}{2} - \tan^{-1} x$:

$$\begin{aligned}
5\tan^{-1} x + 3\left(\frac{\pi}{2} - \tan^{-1} x\right) &= 2\pi \\
5\tan^{-1} x + \frac{3\pi}{2} - 3\tan^{-1} x &= 2\pi \\
2\tan^{-1} x &= 2\pi - \frac{3\pi}{2} \\
2\tan^{-1} x &= \frac{\pi}{2} \\
\tan^{-1} x &= \frac{\pi}{4} \\
x &= \tan\left(\frac{\pi}{4}\right) \\
x &= 1
\end{aligned}$$

(viii) For $x > 0$, $\tan^{-1}\left(\frac{1}{x}\right) = \cot^{-1} x$:

$$\begin{aligned}
\cot^{-1} x &= \cot^{-1}\left(\frac{4}{x}\right) \\
x &= \frac{4}{x} \\
x^2 &= 4 \\
x &= 2 \quad (\text{since } x > 0)
\end{aligned}$$

Final Answer

(i) $x = 1$ (ii) $x = \frac{1}{2}$ (iii) No real solution (iv) $x = \pm\frac{\sqrt{3}}{2}$

(v) $x = \frac{1}{2}$ (vi) $x = \frac{\sqrt{3}}{2}$ (vii) $x = 1$ (viii) $x = 2$

Common Mistake: In (iii), forgetting that $\cos^{-1} x \in [0, \pi]$ and incorrectly writing $x = \cos(-\pi/6) = \sqrt{3}/2$ is a standard trap.

Question 7

If $\sin\left(\sin^{-1}\dfrac{2}{3} + \cos^{-1} x\right) = 1$, then find the value of x.

TYPE 12 Solving Inverse Trigonometric Equations

Given: The equation $\sin\left(\sin^{-1}\dfrac{2}{3} + \cos^{-1} x\right) = 1$.

To Find: The value of x.

Concept / Formula Used: $\sin\theta = 1 \implies \theta = \dfrac{\pi}{2} \qquad \sin^{-1} a + \cos^{-1} a = \dfrac{\pi}{2}$

Solution:

Taking sine inverse on both sides:

$$\begin{aligned} \sin^{-1}\frac{2}{3} + \cos^{-1} x &= \sin^{-1}(1) \\ \sin^{-1}\frac{2}{3} + \cos^{-1} x &= \frac{\pi}{2} \\ \cos^{-1} x &= \frac{\pi}{2} - \sin^{-1}\frac{2}{3} \end{aligned}$$

Using the standard identity $\cos^{-1} a = \dfrac{\pi}{2} - \sin^{-1} a$:

$$\begin{aligned} \cos^{-1} x &= \cos^{-1}\frac{2}{3} \\ x &= \frac{2}{3} \end{aligned}$$

Final Answer

$x = \dfrac{2}{3}$

Question 8

Find the values of:

(i) $\tan^{-1}\dfrac{1}{2} + \tan^{-1}\dfrac{1}{3}$

(ii) $\cot^{-1} 2 + \cot^{-1} 3$

TYPE 13 Addition and Subtraction Formulas for Inverse Trigonometric Functions

To Find: The value of each sum.

Concept / Formula Used: $\tan^{-1} x + \tan^{-1} y = \tan^{-1}\left(\frac{x+y}{1-xy}\right)$ for $xy < 1$

Solution:

(i) Here $x = \frac{1}{2}$ and $y = \frac{1}{3}$. Check product: $xy = \frac{1}{2} \times \frac{1}{3} = \frac{1}{6} < 1$. Using the addition identity:

$$\begin{aligned}
\tan^{-1}\frac{1}{2} + \tan^{-1}\frac{1}{3} &= \tan^{-1}\left(\frac{\frac{1}{2}+\frac{1}{3}}{1-\frac{1}{2}\cdot\frac{1}{3}}\right) \\
&= \tan^{-1}\left(\frac{\frac{5}{6}}{1-\frac{1}{6}}\right) \\
&= \tan^{-1}\left(\frac{\frac{5}{6}}{\frac{5}{6}}\right) \\
&= \tan^{-1}(1) \\
&= \frac{\pi}{4}
\end{aligned}$$

(ii) Using $\cot^{-1} a = \tan^{-1}\left(\frac{1}{a}\right)$ for $a > 0$:

$$\begin{aligned}
\cot^{-1} 2 &= \tan^{-1}\frac{1}{2} \\
\cot^{-1} 3 &= \tan^{-1}\frac{1}{3} \\
\cot^{-1} 2 + \cot^{-1} 3 &= \tan^{-1}\frac{1}{2} + \tan^{-1}\frac{1}{3} \\
&= \frac{\pi}{4} \quad \text{(from part i)}
\end{aligned}$$

Final Answer

(i) $\frac{\pi}{4}$ (ii) $\frac{\pi}{4}$

Question 9

If $\tan^{-1} x - \tan^{-1} y = \frac{\pi}{4}$, $xy > -1$, then find the value of $x - y - xy$.

TYPE 13 Addition and Subtraction Formulas for Inverse Trigonometric Functions

Given: $\tan^{-1} x - \tan^{-1} y = \frac{\pi}{4}$ with $xy > -1$.

To Find: The value of $x - y - xy$.

Concept / Formula Used: $\tan^{-1} x - \tan^{-1} y = \tan^{-1}\left(\frac{x-y}{1+xy}\right)$ for $xy > -1$

Solution:
Applying the subtraction formula:

$$\begin{aligned}
\tan^{-1}\left(\frac{x-y}{1+xy}\right) &= \frac{\pi}{4} \\
\frac{x-y}{1+xy} &= \tan\left(\frac{\pi}{4}\right) \\
\frac{x-y}{1+xy} &= 1 \\
x-y &= 1+xy \\
x-y-xy &= 1
\end{aligned}$$

Final Answer
$x-y-xy=1$

Question 10

(i) If $\tan^{-1} x = \dfrac{\pi}{10}$ for some $x \in \mathbb{R}$, find the value of $\cot^{-1} x$.
(ii) If $\sin^{-1} x + \sin^{-1} y = \dfrac{\pi}{2}$, then find the value of $\cos^{-1} x + \cos^{-1} y$.
(iii) If $\sec^{-1} x + \sec^{-1} y = \dfrac{2\pi}{3}$, then find the value of $\csc^{-1} x + \csc^{-1} y$.

TYPE 7 Evaluating Expressions with Inverse Trigonometric Functions

★ **Likely Exam Question**

To Find: The required values using complementary angle identities.

Concept / Formula Used: $\tan^{-1} x + \cot^{-1} x = \dfrac{\pi}{2}$ $\quad\sin^{-1} x + \cos^{-1} x = \dfrac{\pi}{2}$ $\quad\sec^{-1} x + \csc^{-1} x = \dfrac{\pi}{2}$

Solution:

(i) Using $\tan^{-1} x + \cot^{-1} x = \dfrac{\pi}{2}$:

$$\begin{aligned}
\frac{\pi}{10} + \cot^{-1} x &= \frac{\pi}{2} \\
\cot^{-1} x &= \frac{\pi}{2} - \frac{\pi}{10} \\
&= \frac{5\pi - \pi}{10} \\
&= \frac{4\pi}{10} \\
&= \frac{2\pi}{5}
\end{aligned}$$

(ii) Using $\sin^{-1} t = \dfrac{\pi}{2} - \cos^{-1} t$ for both x and y:

$$\begin{aligned}\left(\frac{\pi}{2} - \cos^{-1} x\right) + \left(\frac{\pi}{2} - \cos^{-1} y\right) &= \frac{\pi}{2}\\ \pi - (\cos^{-1} x + \cos^{-1} y) &= \frac{\pi}{2}\\ \cos^{-1} x + \cos^{-1} y &= \pi - \frac{\pi}{2}\\ &= \frac{\pi}{2}\end{aligned}$$

(iii) Using $\sec^{-1} t = \dfrac{\pi}{2} - \csc^{-1} t$ for both x and y:

$$\begin{aligned}\left(\frac{\pi}{2} - \csc^{-1} x\right) + \left(\frac{\pi}{2} - \csc^{-1} y\right) &= \frac{2\pi}{3}\\ \pi - (\csc^{-1} x + \csc^{-1} y) &= \frac{2\pi}{3}\\ \csc^{-1} x + \csc^{-1} y &= \pi - \frac{2\pi}{3}\\ &= \frac{\pi}{3}\end{aligned}$$

Final Answer

(i) $\dfrac{2\pi}{5}$ (ii) $\dfrac{\pi}{2}$ (iii) $\dfrac{\pi}{3}$

Question 11

If $\cos^{-1} x + \cos^{-1} y + \cos^{-1} z = 3\pi$, then find the value of

$$x^{100} + y^{100} + z^{100} - \frac{9}{x^{101} + y^{101} + z^{101}}$$

TYPE 5 **Extrema of Inverse Trigonometric Functions**

★ **Likely Exam Question**

Given: $\cos^{-1} x + \cos^{-1} y + \cos^{-1} z = 3\pi$.

To Find: The value of $x^{100} + y^{100} + z^{100} - \dfrac{9}{x^{101} + y^{101} + z^{101}}$.

Concept / Formula Used: $\cos^{-1} t$ $[0, \pi]$ $\cos^{-1} t$ π $t = -1$

Solution:

Since $0 \le \cos^{-1} t \le \pi$ for all $t \in [-1, 1]$:

$$\cos^{-1} x + \cos^{-1} y + \cos^{-1} z \le \pi + \pi + \pi = 3\pi$$

The equality holds if and only if each term achieves its maximum value:

$$\cos^{-1} x = \pi \implies x = \cos \pi = -1$$
$$\cos^{-1} y = \pi \implies y = \cos \pi = -1$$
$$\cos^{-1} z = \pi \implies z = \cos \pi = -1$$

Now substituting $x = -1, y = -1, z = -1$ into the given expression:

$$\begin{aligned} x^{100} + y^{100} + z^{100} &= (-1)^{100} + (-1)^{100} + (-1)^{100} \\ &= 1 + 1 + 1 = 3 \\ x^{101} + y^{101} + z^{101} &= (-1)^{101} + (-1)^{101} + (-1)^{101} \\ &= -1 - 1 - 1 = -3 \\ \therefore \text{Expression} &= 3 - \frac{9}{-3} \\ &= 3 - (-3) \\ &= 3 + 3 \\ &= 6 \end{aligned}$$

Final Answer

6

Question 12

Prove the following:

(i) $\tan^{-1}\left(\dfrac{1+x}{1-x}\right) = \dfrac{\pi}{4} + \tan^{-1} x, \quad x < 1$

(ii) $\tan^{-1} x + \cot^{-1}(x + 1) = \tan^{-1}(x^2 + x + 1)$

(iii) $\cos\left[\tan^{-1}\{\cot(\sin^{-1} x)\}\right] = x$

TYPE 13 **Addition and Subtraction Formulas for Inverse Trigonometric Functions**

To Prove: (i) $\tan^{-1}\left(\dfrac{1+x}{1-x}\right) = \dfrac{\pi}{4} + \tan^{-1} x$

(ii) $\tan^{-1} x + \cot^{-1}(x + 1) = \tan^{-1}(x^2 + x + 1)$

(iii) $\cos\left[\tan^{-1}\{\cot(\sin^{-1} x)\}\right] = x$

Concept / Formula Used: $\tan^{-1}\left(\dfrac{a+b}{1-ab}\right) = \tan^{-1} a + \tan^{-1} b \qquad \cot^{-1} y = \tan^{-1}\left(\dfrac{1}{y}\right)$ for $y > 0$

Solution:

(i) Starting with the Left Hand Side (LHS):

$$\text{LHS} = \tan^{-1}\left(\frac{1+x}{1-1\cdot x}\right)$$

Using the identity $\tan^{-1}\left(\frac{a+b}{1-ab}\right) = \tan^{-1} a + \tan^{-1} b$ with $a = 1$ and $b = x$:

$$\begin{aligned} \text{LHS} &= \tan^{-1}(1) + \tan^{-1} x \\ &= \frac{\pi}{4} + \tan^{-1} x = \text{RHS} \end{aligned}$$

(ii) Convert $\cot^{-1}(x+1)$ to $\tan^{-1}\left(\frac{1}{x+1}\right)$:

$$\text{LHS} = \tan^{-1} x + \tan^{-1}\left(\frac{1}{x+1}\right)$$

Apply addition formula for $\tan^{-1}$:

$$\begin{aligned} \text{LHS} &= \tan^{-1}\left(\frac{x + \frac{1}{x+1}}{1 - x \cdot \frac{1}{x+1}}\right) \\ &= \tan^{-1}\left(\frac{\frac{x(x+1)+1}{x+1}}{\frac{(x+1)-x}{x+1}}\right) \\ &= \tan^{-1}\left(\frac{x^2+x+1}{1}\right) \\ &= \tan^{-1}(x^2+x+1) = \text{RHS} \end{aligned}$$

(iii) Work from the innermost term outwards:

$$\begin{aligned} \text{Let } \theta = \sin^{-1} x &\implies \sin\theta = x \\ \cos\theta &= \sqrt{1-x^2} \\ \cot(\sin^{-1} x) = \cot\theta &= \frac{\cos\theta}{\sin\theta} = \frac{\sqrt{1-x^2}}{x} \end{aligned}$$

Now substitute this into the middle term:

$$\text{Let } \phi = \tan^{-1}\left(\frac{\sqrt{1-x^2}}{x}\right) \implies \tan\phi = \frac{\sqrt{1-x^2}}{x}$$

Draw a right triangle for ϕ:

$$\begin{aligned} \text{Perpendicular} &= \sqrt{1-x^2}, \quad \text{Base} = x \\ \text{Hypotenuse} &= \sqrt{(\sqrt{1-x^2})^2 + x^2} = \sqrt{1-x^2+x^2} = 1 \\ \cos\phi &= \frac{\text{Base}}{\text{Hypotenuse}} = \frac{x}{1} = x \end{aligned}$$

Therefore:

$$\text{LHS} = \cos\phi = x = \text{RHS}$$

Hence Proved

LHS = RHS in all subparts. Hence proved.

Question 13

Prove the following (for suitable values of x and y):

(i) $\tan^{-1}\sqrt{x} + \tan^{-1}\sqrt{y} = \tan^{-1}\dfrac{\sqrt{x}+\sqrt{y}}{1-\sqrt{xy}}$

(ii) $\tan^{-1}\dfrac{x+\sqrt{x}}{1-x^{3/2}} = \tan^{-1}x + \tan^{-1}\sqrt{x}$

TYPE 13 **Addition and Subtraction Formulas for Inverse Trigonometric Functions**

To Prove: (i) $\tan^{-1}\sqrt{x} + \tan^{-1}\sqrt{y} = \tan^{-1}\dfrac{\sqrt{x}+\sqrt{y}}{1-\sqrt{xy}}$

(ii) $\tan^{-1}\dfrac{x+\sqrt{x}}{1-x^{3/2}} = \tan^{-1}x + \tan^{-1}\sqrt{x}$

Concept / Formula Used: $\tan^{-1}A + \tan^{-1}B = \tan^{-1}\left(\dfrac{A+B}{1-AB}\right)$

Solution:

(i) Apply addition formula with $A = \sqrt{x}$ and $B = \sqrt{y}$:

$$\begin{aligned} \text{LHS} &= \tan^{-1}\sqrt{x} + \tan^{-1}\sqrt{y} \\ &= \tan^{-1}\left(\frac{\sqrt{x}+\sqrt{y}}{1-\sqrt{x}\cdot\sqrt{y}}\right) \\ &= \tan^{-1}\left(\frac{\sqrt{x}+\sqrt{y}}{1-\sqrt{xy}}\right) = \text{RHS} \end{aligned}$$

(ii) Rewrite the denominator of the LHS:

$$x^{3/2} = x \cdot x^{1/2} = x\sqrt{x}$$

$$\text{LHS} = \tan^{-1}\left(\frac{x+\sqrt{x}}{1-x\sqrt{x}}\right)$$

Using the identity $\tan^{-1}\left(\dfrac{A+B}{1-AB}\right) = \tan^{-1}A + \tan^{-1}B$ with $A = x$ and $B = \sqrt{x}$:

$$\text{LHS} = \tan^{-1}x + \tan^{-1}\sqrt{x} = \text{RHS}$$

Hence Proved

LHS = RHS for both parts. Hence proved.

Question 14

Write the following functions in simplest form:

(i) $\tan^{-1}\left(\sqrt{\dfrac{1-\cos x}{1+\cos x}}\right)$

(ii) $\tan^{-1}\left(\dfrac{\cos x-\sin x}{\cos x+\sin x}\right)$

(iii) $\tan^{-1}\left(\dfrac{2\sqrt{x}}{1-x}\right)$

(iv) $\sin^{-1}\left(\sqrt{\dfrac{x}{1+x}}\right)$

(v) $\tan^{-1}\left(\sqrt{\dfrac{1-x}{1+x}}\right)$

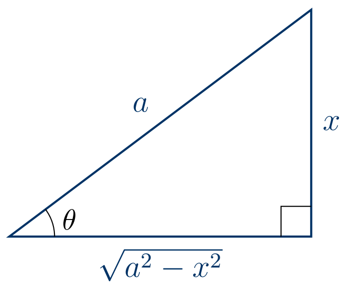
(vi) $\cos^{-1}(1-2x^2)$

(vii) $\sec^{-1}\left(\dfrac{1}{2x^2-1}\right)$

(viii) $\tan^{-1}\left(\dfrac{x}{\sqrt{a^2-x^2}}\right)$

TYPE 11 **Simplifying Inverse Trigonometric Expressions**

★ Likely Exam Question



Right triangle for substitution $x = a\sin\theta$ *in* $\tan^{-1}\left(\frac{x}{\sqrt{a^2-x^2}}\right)$

To Find: The simplest simplified form of each expression.

Concept / Formula Used: $1-\cos x = 2\sin^2\dfrac{x}{2}$, $1+\cos x = 2\cos^2\dfrac{x}{2}$ $\tan\left(\dfrac{\pi}{4}-x\right) = \dfrac{1-\tan x}{1+\tan x}$ $2\tan^{-1}u = \tan^{-1}\left(\dfrac{2u}{1-u^2}\right)$ $\cos 2\theta = 1-2\sin^2\theta = 2\cos^2\theta - 1$

Solution:

(i) Using standard half-angle forms:

$$\sqrt{\frac{1-\cos x}{1+\cos x}} = \sqrt{\frac{2\sin^2\frac{x}{2}}{2\cos^2\frac{x}{2}}} = \sqrt{\tan^2\frac{x}{2}} = \tan\frac{x}{2}$$
$$\tan^{-1}\left(\sqrt{\frac{1-\cos x}{1+\cos x}}\right) = \tan^{-1}\left(\tan\frac{x}{2}\right) = \frac{x}{2}$$

(ii) Divide numerator and denominator by $\cos x$:

$$\frac{\cos x - \sin x}{\cos x + \sin x} = \frac{1-\tan x}{1+\tan x} = \tan\left(\frac{\pi}{4} - x\right)$$
$$\tan^{-1}\left(\frac{\cos x - \sin x}{\cos x + \sin x}\right) = \tan^{-1}\left(\tan\left(\frac{\pi}{4} - x\right)\right) = \frac{\pi}{4} - x$$

(iii) Let $\sqrt{x} = \tan\theta \implies \theta = \tan^{-1}\sqrt{x}$.

$$\frac{2\sqrt{x}}{1-x} = \frac{2\tan\theta}{1-\tan^2\theta} = \tan 2\theta$$
$$\tan^{-1}\left(\frac{2\sqrt{x}}{1-x}\right) = \tan^{-1}(\tan 2\theta) = 2\theta = 2\tan^{-1}\sqrt{x}$$

(iv) Let $x = \tan^2\theta \implies \tan\theta = \sqrt{x} \implies \theta = \tan^{-1}\sqrt{x}$.

$$\sqrt{\frac{x}{1+x}} = \sqrt{\frac{\tan^2\theta}{1+\tan^2\theta}} = \sqrt{\frac{\tan^2\theta}{\sec^2\theta}} = \sin\theta$$
$$\sin^{-1}\left(\sqrt{\frac{x}{1+x}}\right) = \sin^{-1}(\sin\theta) = \theta = \tan^{-1}\sqrt{x}$$

(v) Let $x = \cos 2\theta \implies 2\theta = \cos^{-1}x \implies \theta = \frac{1}{2}\cos^{-1}x$.

$$\sqrt{\frac{1-x}{1+x}} = \sqrt{\frac{1-\cos 2\theta}{1+\cos 2\theta}} = \sqrt{\frac{2\sin^2\theta}{2\cos^2\theta}} = \tan\theta$$
$$\tan^{-1}\left(\sqrt{\frac{1-x}{1+x}}\right) = \tan^{-1}(\tan\theta) = \theta = \frac{1}{2}\cos^{-1}x$$

(vi) Let $x = \sin\theta \implies \theta = \sin^{-1}x$.

$$1-2x^2 = 1-2\sin^2\theta = \cos 2\theta$$
$$\cos^{-1}(1-2x^2) = \cos^{-1}(\cos 2\theta) = 2\theta = 2\sin^{-1}x$$

(vii) Convert $\sec^{-1}$ to $\cos^{-1}$:

$$\sec^{-1}\left(\frac{1}{2x^2-1}\right) = \cos^{-1}(2x^2-1)$$

Let $x = \cos\theta \implies \theta = \cos^{-1}x$

$$2x^2-1 = 2\cos^2\theta - 1 = \cos 2\theta$$
$$\cos^{-1}(2x^2-1) = \cos^{-1}(\cos 2\theta) = 2\theta = 2\cos^{-1}x$$

(viii) Let $x = a\sin\theta \implies \sin\theta = \frac{x}{a} \implies \theta = \sin^{-1}\left(\frac{x}{a}\right)$.

$$\sqrt{a^2 - x^2} = \sqrt{a^2 - a^2\sin^2\theta} = a\cos\theta$$
$$\frac{x}{\sqrt{a^2 - x^2}} = \frac{a\sin\theta}{a\cos\theta} = \tan\theta$$
$$\tan^{-1}\left(\frac{x}{\sqrt{a^2 - x^2}}\right) = \tan^{-1}(\tan\theta) = \theta = \sin^{-1}\left(\frac{x}{a}\right)$$

Final Answer

(i) $\frac{x}{2}$ (ii) $\frac{\pi}{4} - x$ (iii) $2\tan^{-1}\sqrt{x}$ (iv) $\tan^{-1}\sqrt{x}$
(v) $\frac{1}{2}\cos^{-1}x$ (vi) $2\sin^{-1}x$ (vii) $2\cos^{-1}x$ (viii) $\sin^{-1}\left(\frac{x}{a}\right)$

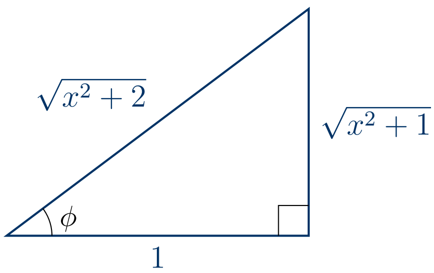
Question 15

Prove that:

$$\sin\left(\cot^{-1}\left(\cos\left(\tan^{-1}x\right)\right)\right) = \sqrt{\frac{x^2+1}{x^2+2}}$$

TYPE 8 Evaluating Nested Inverse Trigonometric Functions

★ **Likely Exam Question**



Right triangle transformations for step-by-step nested inverse functions

To Prove: $\sin\left(\cot^{-1}\left(\cos\left(\tan^{-1}x\right)\right)\right) = \sqrt{\frac{x^2+1}{x^2+2}}$

Concept / Formula Used: $\tan\theta = x$ $\cos\theta = \frac{1}{\sqrt{1+x^2}}$ $\cot\phi = k$ $\sin\phi = \frac{1}{\sqrt{1+k^2}}$

Solution:

Work step-by-step from the innermost function to the outermost:

Step 1: Let $\theta = \tan^{-1} x \implies \tan\theta = x$.

$$\text{Perpendicular} = x, \quad \text{Base} = 1, \quad \text{Hypotenuse} = \sqrt{1+x^2}$$
$$\cos(\tan^{-1} x) = \cos\theta = \frac{1}{\sqrt{1+x^2}}$$

Step 2: Substitute this back into the expression:

$$\text{Expression} = \sin\left(\cot^{-1}\left(\frac{1}{\sqrt{1+x^2}}\right)\right)$$

Let $\phi = \cot^{-1}\left(\frac{1}{\sqrt{1+x^2}}\right) \implies \cot\phi = \frac{1}{\sqrt{1+x^2}}$.

$$\text{Base} = 1, \quad \text{Perpendicular} = \sqrt{1+x^2}$$
$$\begin{aligned}\text{Hypotenuse} &= \sqrt{\text{Base}^2 + \text{Perpendicular}^2} \\ &= \sqrt{1^2 + \left(\sqrt{1+x^2}\right)^2} \\ &= \sqrt{1+1+x^2} \\ &= \sqrt{x^2+2}\end{aligned}$$

Step 3: Evaluate $\sin\phi$:

$$\sin\phi = \frac{\text{Perpendicular}}{\text{Hypotenuse}} = \frac{\sqrt{1+x^2}}{\sqrt{x^2+2}} = \sqrt{\frac{x^2+1}{x^2+2}}$$

Hence Proved

LHS = RHS = $\sqrt{\frac{x^2+1}{x^2+2}}$. Hence proved.

Question 16

Find the value of

$$\tan\frac{1}{2}\left(\sin^{-1}\frac{2x}{1+x^2} + \cos^{-1}\frac{1-y^2}{1+y^2}\right), \quad |x| < 1,\ y > 0,\ xy < 1$$

TYPE 13 **Addition and Subtraction Formulas for Inverse Trigonometric Functions**

★ **Likely Exam Question**

Given: Expression $\tan\frac{1}{2}\left(\sin^{-1}\frac{2x}{1+x^2} + \cos^{-1}\frac{1-y^2}{1+y^2}\right)$ with $|x| < 1, y > 0, xy < 1$.

To Find: The value of the expression in terms of x and y.

Concept / Formula Used: $\sin^{-1}\left(\frac{2x}{1+x^2}\right) = 2\tan^{-1}x$ for $|x| < 1$ $\cos^{-1}\left(\frac{1-y^2}{1+y^2}\right) = 2\tan^{-1}y$ for $y > 0$ $\tan^{-1}x + \tan^{-1}y = \tan^{-1}\left(\frac{x+y}{1-xy}\right)$ for $xy < 1$

Solution:

Substitute the standard 2-angle identities into the expression:

$$\sin^{-1}\left(\frac{2x}{1+x^2}\right) = 2\tan^{-1}x$$
$$\cos^{-1}\left(\frac{1-y^2}{1+y^2}\right) = 2\tan^{-1}y$$

Substituting these into the given expression:

$$\begin{aligned}\tan\left[\frac{1}{2}\left(2\tan^{-1}x + 2\tan^{-1}y\right)\right] &= \tan\left[\frac{1}{2}\cdot 2\left(\tan^{-1}x + \tan^{-1}y\right)\right] \\ &= \tan\left(\tan^{-1}x + \tan^{-1}y\right)\end{aligned}$$

Now apply the addition formula $\tan^{-1}x + \tan^{-1}y = \tan^{-1}\left(\frac{x+y}{1-xy}\right)$:

$$\begin{aligned}&= \tan\left(\tan^{-1}\left(\frac{x+y}{1-xy}\right)\right) \\ &= \frac{x+y}{1-xy}\end{aligned}$$

Final Answer

$\frac{x+y}{1-xy}$

Question 17

Prove the following:

(i) $\tan^{-1}\frac{1}{2} + \tan^{-1}\frac{2}{11} = \tan^{-1}\frac{3}{4}$

(ii) $\tan^{-1}\frac{1}{7} + \tan^{-1}\frac{1}{13} = \tan^{-1}\frac{2}{9}$

(iii) $\tan^{-1}2 - \tan^{-1}1 = \tan^{-1}\frac{1}{3}$

(iv) $\tan^{-1}1 + \tan^{-1}\frac{1}{2} + \tan^{-1}\frac{1}{3} = \frac{\pi}{2}$

TYPE 13 Addition and Subtraction Formulas for Inverse Trigonometric Functions

★ **Likely Exam Question**

To Prove: (i) $\tan^{-1}\frac{1}{2}+\tan^{-1}\frac{2}{11}=\tan^{-1}\frac{3}{4}$
(ii) $\tan^{-1}\frac{1}{7}+\tan^{-1}\frac{1}{13}=\tan^{-1}\frac{2}{9}$
(iii) $\tan^{-1}2-\tan^{-1}1=\tan^{-1}\frac{1}{3}$
(iv) $\tan^{-1}1+\tan^{-1}\frac{1}{2}+\tan^{-1}\frac{1}{3}=\frac{\pi}{2}$

Concept / Formula Used: $\tan^{-1}x+\tan^{-1}y=\tan^{-1}\left(\frac{x+y}{1-xy}\right)$ $\tan^{-1}x-\tan^{-1}y=\tan^{-1}\left(\frac{x-y}{1+xy}\right)$

Solution:

(i) Apply addition formula with $x=\frac{1}{2}, y=\frac{2}{11}$:

$$\begin{aligned}\text{LHS}&=\tan^{-1}\left(\frac{\frac{1}{2}+\frac{2}{11}}{1-\frac{1}{2}\cdot\frac{2}{11}}\right)\\&=\tan^{-1}\left(\frac{\frac{11+4}{22}}{1-\frac{2}{22}}\right)\\&=\tan^{-1}\left(\frac{\frac{15}{22}}{\frac{20}{22}}\right)\\&=\tan^{-1}\left(\frac{15}{20}\right)\\&=\tan^{-1}\left(\frac{3}{4}\right)=\text{RHS}\end{aligned}$$

(ii) Apply addition formula with $x=\frac{1}{7}, y=\frac{1}{13}$:

$$\begin{aligned}\text{LHS}&=\tan^{-1}\left(\frac{\frac{1}{7}+\frac{1}{13}}{1-\frac{1}{7}\cdot\frac{1}{13}}\right)\\&=\tan^{-1}\left(\frac{\frac{13+7}{91}}{1-\frac{1}{91}}\right)\\&=\tan^{-1}\left(\frac{\frac{20}{91}}{\frac{90}{91}}\right)\\&=\tan^{-1}\left(\frac{20}{90}\right)\\&=\tan^{-1}\left(\frac{2}{9}\right)=\text{RHS}\end{aligned}$$

(iii) Apply subtraction formula with $x = 2, y = 1$:

$$\begin{aligned} \text{LHS} &= \tan^{-1}\left(\frac{2-1}{1+2\cdot 1}\right) \\ &= \tan^{-1}\left(\frac{1}{1+2}\right) \\ &= \tan^{-1}\left(\frac{1}{3}\right) = \text{RHS} \end{aligned}$$

(iv) Combine terms stepwise:

$$\begin{aligned} \text{LHS} &= \tan^{-1}(1) + \left(\tan^{-1}\frac{1}{2} + \tan^{-1}\frac{1}{3}\right) \\ &= \frac{\pi}{4} + \tan^{-1}\left(\frac{\frac{1}{2}+\frac{1}{3}}{1-\frac{1}{2}\cdot\frac{1}{3}}\right) \\ &= \frac{\pi}{4} + \tan^{-1}\left(\frac{\frac{5}{6}}{\frac{5}{6}}\right) \\ &= \frac{\pi}{4} + \tan^{-1}(1) \\ &= \frac{\pi}{4} + \frac{\pi}{4} \\ &= \frac{\pi}{2} = \text{RHS} \end{aligned}$$

Hence Proved

LHS = RHS for all four parts. Hence proved.

Question 17 (v)

Prove that $2\tan^{-1}\frac{1}{3} + \tan^{-1}\frac{1}{7} = \frac{\pi}{4}$.

TYPE 13 **Addition and Subtraction Formulas for Inverse Trigonometric Functions**

To Prove: $2\tan^{-1}\frac{1}{3} + \tan^{-1}\frac{1}{7} = \frac{\pi}{4}$

Concept / Formula Used: $2\tan^{-1}x = \tan^{-1}\left(\frac{2x}{1-x^2}\right)$ $\tan^{-1}x + \tan^{-1}y = \tan^{-1}\left(\frac{x+y}{1-xy}\right)$

Solution:

Using the standard identity $2\tan^{-1} x = \tan^{-1}\left(\frac{2x}{1-x^2}\right)$ for $x = \frac{1}{3}$:

$$
\begin{aligned}
2\tan^{-1}\frac{1}{3} &= \tan^{-1}\left(\frac{2\times\frac{1}{3}}{1-\left(\frac{1}{3}\right)^2}\right)\\
&= \tan^{-1}\left(\frac{\frac{2}{3}}{1-\frac{1}{9}}\right)\\
&= \tan^{-1}\left(\frac{\frac{2}{3}}{\frac{8}{9}}\right)\\
&= \tan^{-1}\left(\frac{2}{3}\times\frac{9}{8}\right)\\
&= \tan^{-1}\frac{3}{4}
\end{aligned}
$$

Substitute this value into the left-hand side:

$$\text{LHS} = \tan^{-1}\frac{3}{4} + \tan^{-1}\frac{1}{7}$$

Using the addition identity $\tan^{-1} x + \tan^{-1} y = \tan^{-1}\left(\frac{x+y}{1-xy}\right)$:

$$
\begin{aligned}
\text{LHS} &= \tan^{-1}\left(\frac{\frac{3}{4}+\frac{1}{7}}{1-\frac{3}{4}\times\frac{1}{7}}\right)\\
&= \tan^{-1}\left(\frac{\frac{21+4}{28}}{1-\frac{3}{28}}\right)\\
&= \tan^{-1}\left(\frac{\frac{25}{28}}{\frac{25}{28}}\right)\\
&= \tan^{-1}(1)\\
&= \frac{\pi}{4} = \text{RHS}
\end{aligned}
$$

Hence Proved

LHS = RHS, hence proved.

Question 17 (vi)

Prove that $\tan^{-1}\frac{3}{4} + \tan^{-1}\frac{3}{5} - \tan^{-1}\frac{8}{19} = \frac{\pi}{4}$.

TYPE 13 Addition and Subtraction Formulas for Inverse Trigonometric Functions

To Prove: $\tan^{-1}\frac{3}{4} + \tan^{-1}\frac{3}{5} - \tan^{-1}\frac{8}{19} = \frac{\pi}{4}$

Concept / Formula Used: $\tan^{-1}x + \tan^{-1}y = \tan^{-1}\left(\dfrac{x+y}{1-xy}\right)$ $\tan^{-1}x - \tan^{-1}y = \tan^{-1}\left(\dfrac{x-y}{1+xy}\right)$

Solution:

First, combine the first two terms using $\tan^{-1}x + \tan^{-1}y = \tan^{-1}\left(\dfrac{x+y}{1-xy}\right)$:

$$\begin{aligned}
\tan^{-1}\frac{3}{4} + \tan^{-1}\frac{3}{5} &= \tan^{-1}\left(\frac{\frac{3}{4}+\frac{3}{5}}{1-\frac{3}{4}\times\frac{3}{5}}\right)\\
&= \tan^{-1}\left(\frac{\frac{15+12}{20}}{1-\frac{9}{20}}\right)\\
&= \tan^{-1}\left(\frac{\frac{27}{20}}{\frac{11}{20}}\right)\\
&= \tan^{-1}\frac{27}{11}
\end{aligned}$$

Substitute this result back into the left-hand side:

$$\text{LHS} = \tan^{-1}\frac{27}{11} - \tan^{-1}\frac{8}{19}$$

Using the subtraction identity $\tan^{-1}x - \tan^{-1}y = \tan^{-1}\left(\dfrac{x-y}{1+xy}\right)$:

$$\begin{aligned}
\text{LHS} &= \tan^{-1}\left(\frac{\frac{27}{11}-\frac{8}{19}}{1+\frac{27}{11}\times\frac{8}{19}}\right)\\
&= \tan^{-1}\left(\frac{\frac{27\times19-8\times11}{209}}{\frac{209+216}{209}}\right)\\
&= \tan^{-1}\left(\frac{513-88}{425}\right)\\
&= \tan^{-1}\left(\frac{425}{425}\right)\\
&= \tan^{-1}(1)\\
&= \frac{\pi}{4} = \text{RHS}
\end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

Question 17 (vii)

Prove that $\cot^{-1}1 + \cot^{-1}2 + \cot^{-1}3 = \dfrac{\pi}{2}$.

TYPE 13 **Addition and Subtraction Formulas for Inverse Trigonometric Functions**

To Prove: $\cot^{-1} 1 + \cot^{-1} 2 + \cot^{-1} 3 = \frac{\pi}{2}$

Concept / Formula Used: $\cot^{-1} x = \tan^{-1}\left(\frac{1}{x}\right) \qquad x > 0$

Solution:

Using the standard identity $\cot^{-1} x = \tan^{-1}\left(\frac{1}{x}\right)$ for $x > 0$:

$$\cot^{-1} 1 = \tan^{-1} 1$$
$$\cot^{-1} 2 = \tan^{-1} \frac{1}{2}$$
$$\cot^{-1} 3 = \tan^{-1} \frac{1}{3}$$

Rewrite the left-hand side:

$$\text{LHS} = \tan^{-1} 1 + \tan^{-1} \frac{1}{2} + \tan^{-1} \frac{1}{3}$$

Combine $\tan^{-1} \frac{1}{2} + \tan^{-1} \frac{1}{3}$ using $\tan^{-1} x + \tan^{-1} y = \tan^{-1}\left(\frac{x+y}{1-xy}\right)$:

$$\begin{aligned}
\tan^{-1} \frac{1}{2} + \tan^{-1} \frac{1}{3} &= \tan^{-1}\left(\frac{\frac{1}{2}+\frac{1}{3}}{1-\frac{1}{2}\times\frac{1}{3}}\right) \\
&= \tan^{-1}\left(\frac{\frac{5}{6}}{1-\frac{1}{6}}\right) \\
&= \tan^{-1}\left(\frac{\frac{5}{6}}{\frac{5}{6}}\right) \\
&= \tan^{-1}(1) \\
&= \frac{\pi}{4}
\end{aligned}$$

Substitute this result back into the expression:

$$\begin{aligned}
\text{LHS} &= \tan^{-1} 1 + \frac{\pi}{4} \\
&= \frac{\pi}{4} + \frac{\pi}{4} \\
&= \frac{\pi}{2} = \text{RHS}
\end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

Question 17 (viii)

Prove that $\tan^{-1} 2 + \tan^{-1} 3 = \frac{3\pi}{4}$.

TYPE 13 Addition and Subtraction Formulas for Inverse Trigonometric Functions

★ **Likely Exam Question**

To Prove: $\tan^{-1} 2 + \tan^{-1} 3 = \dfrac{3\pi}{4}$

Concept / Formula Used: $x > 0, y > 0 \quad xy > 1 \quad \tan^{-1} x + \tan^{-1} y = \pi + \tan^{-1}\left(\dfrac{x+y}{1-xy}\right)$

Solution:

Here $x = 2 > 0$ and $y = 3 > 0$, so $xy = 2 \times 3 = 6 > 1$. Using the standard identity $\tan^{-1} x + \tan^{-1} y = \pi + \tan^{-1}\left(\dfrac{x+y}{1-xy}\right)$:

$$\begin{aligned}
\text{LHS} &= \pi + \tan^{-1}\left(\frac{2+3}{1-2\times 3}\right) \\
&= \pi + \tan^{-1}\left(\frac{5}{1-6}\right) \\
&= \pi + \tan^{-1}\left(\frac{5}{-5}\right) \\
&= \pi + \tan^{-1}(-1) \\
&= \pi - \frac{\pi}{4} \\
&= \frac{3\pi}{4} = \text{RHS}
\end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

Common Mistake: Applying $\tan^{-1} x + \tan^{-1} y = \tan^{-1}\left(\frac{x+y}{1-xy}\right)$ without adding π when $xy > 1$ yields $-\frac{\pi}{4}$, which is incorrect because both terms are positive angles.

Question 18

If $\tan^{-1}\left(\dfrac{x-1}{x+1}\right) + \tan^{-1}\left(\dfrac{2x-1}{2x+1}\right) = \tan^{-1}\left(\dfrac{23}{36}\right)$, then prove that $24x^2 - 23x - 12 = 0$.

TYPE 13 Addition and Subtraction Formulas for Inverse Trigonometric Functions

★ **Likely Exam Question**

To Prove: $24x^2-23x-12=0$ given $\tan^{-1}\left(\frac{x-1}{x+1}\right)+\tan^{-1}\left(\frac{2x-1}{2x+1}\right)=\tan^{-1}\left(\frac{23}{36}\right)$

Concept / Formula Used: $\tan^{-1}a+\tan^{-1}b=\tan^{-1}\left(\frac{a+b}{1-ab}\right)$

Solution:

Let $a=\frac{x-1}{x+1}$ and $b=\frac{2x-1}{2x+1}$. Applying the identity $\tan^{-1}a+\tan^{-1}b=\tan^{-1}\left(\frac{a+b}{1-ab}\right)$:

$$\tan^{-1}\left(\frac{\frac{x-1}{x+1}+\frac{2x-1}{2x+1}}{1-\frac{x-1}{x+1}\cdot\frac{2x-1}{2x+1}}\right)=\tan^{-1}\left(\frac{23}{36}\right)$$

Taking tangent on both sides:

$$\frac{\frac{(x-1)(2x+1)+(2x-1)(x+1)}{(x+1)(2x+1)}}{\frac{(x+1)(2x+1)-(x-1)(2x-1)}{(x+1)(2x+1)}}=\frac{23}{36}$$

Cancelling the common denominator $(x+1)(2x+1)$:

$$\frac{(x-1)(2x+1)+(2x-1)(x+1)}{(x+1)(2x+1)-(x-1)(2x-1)}=\frac{23}{36}$$

Expand the numerator:

$$\begin{aligned}\text{Numerator}&=(2x^2+x-2x-1)+(2x^2+2x-x-1)\\&=(2x^2-x-1)+(2x^2+x-1)\\&=4x^2-2\end{aligned}$$

Expand the denominator:

$$\begin{aligned}\text{Denominator}&=(2x^2+3x+1)-(2x^2-3x+1)\\&=2x^2+3x+1-2x^2+3x-1\\&=6x\end{aligned}$$

Substitute these back into the fraction:

$$\frac{4x^2-2}{6x}=\frac{23}{36}$$

Divide numerator and denominator of the left side by 2:

$$\frac{2x^2-1}{3x}=\frac{23}{36}$$

Cross-multiply:

$$\begin{aligned}36(2x^2-1)&=23(3x)\\72x^2-36&=69x\\72x^2-69x-36&=0\end{aligned}$$

Divide the entire equation by 3:

$$24x^2 - 23x - 12 = 0$$

Hence Proved

Derived the equation $24x^2 - 23x - 12 = 0$, hence proved.

Question 19 (i)

Prove that $\sin^{-1}\dfrac{1}{\sqrt{5}} + \sin^{-1}\dfrac{2}{\sqrt{5}} = \dfrac{\pi}{2}$.

TYPE 13 **Addition and Subtraction Formulas for Inverse Trigonometric Functions**

To Prove: $\sin^{-1}\dfrac{1}{\sqrt{5}} + \sin^{-1}\dfrac{2}{\sqrt{5}} = \dfrac{\pi}{2}$

Concept / Formula Used: $\sin^{-1}x + \cos^{-1}x = \dfrac{\pi}{2}$ $\sin^{-1}y = \cos^{-1}\sqrt{1-y^2}$ $y > 0$

Solution:

Let $\theta = \sin^{-1}\dfrac{2}{\sqrt{5}}$. Then $\sin\theta = \dfrac{2}{\sqrt{5}}$. Using $\cos\theta = \sqrt{1-\sin^2\theta}$:

$$\begin{aligned}\cos\theta &= \sqrt{1-\left(\frac{2}{\sqrt{5}}\right)^2}\\ &= \sqrt{1-\frac{4}{5}}\\ &= \sqrt{\frac{1}{5}}\\ &= \frac{1}{\sqrt{5}}\end{aligned}$$

Thus $\theta = \cos^{-1}\dfrac{1}{\sqrt{5}}$, which gives:

$$\sin^{-1}\frac{2}{\sqrt{5}} = \cos^{-1}\frac{1}{\sqrt{5}}$$

Substitute this into the left-hand side:

$$\text{LHS} = \sin^{-1}\frac{1}{\sqrt{5}} + \cos^{-1}\frac{1}{\sqrt{5}}$$

Using the standard result $\sin^{-1}x + \cos^{-1}x = \dfrac{\pi}{2}$ for $x = \dfrac{1}{\sqrt{5}}$:

$$\text{LHS} = \frac{\pi}{2} = \text{RHS}$$

Hence Proved

LHS = RHS, hence proved.

Question 19 (ii)

Prove that $\cos^{-1}\frac{4}{5}+\cos^{-1}\frac{12}{13}=\cos^{-1}\frac{33}{65}$.

TYPE 13 Addition and Subtraction Formulas for Inverse Trigonometric Functions

★ **Likely Exam Question**

To Prove: $\cos^{-1}\frac{4}{5}+\cos^{-1}\frac{12}{13}=\cos^{-1}\frac{33}{65}$

Concept / Formula Used: $\cos^{-1}x+\cos^{-1}y=\cos^{-1}\left(xy-\sqrt{1-x^2}\sqrt{1-y^2}\right)$

Solution:

Let $x=\frac{4}{5}$ and $y=\frac{12}{13}$. Calculate the radical terms separately:

$$\sqrt{1-x^2}=\sqrt{1-\left(\frac{4}{5}\right)^2}=\sqrt{1-\frac{16}{25}}=\sqrt{\frac{9}{25}}=\frac{3}{5}$$

$$\sqrt{1-y^2}=\sqrt{1-\left(\frac{12}{13}\right)^2}=\sqrt{1-\frac{144}{169}}=\sqrt{\frac{25}{169}}=\frac{5}{13}$$

Using the standard identity $\cos^{-1}x+\cos^{-1}y=\cos^{-1}\left(xy-\sqrt{1-x^2}\sqrt{1-y^2}\right)$:

$$\begin{aligned}\text{LHS}&=\cos^{-1}\left(\frac{4}{5}\times\frac{12}{13}-\frac{3}{5}\times\frac{5}{13}\right)\\&=\cos^{-1}\left(\frac{48}{65}-\frac{15}{65}\right)\\&=\cos^{-1}\left(\frac{33}{65}\right)=\text{RHS}\end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

Question 19 (iii)

Prove that $\cos^{-1}\frac{3}{5}+\sin^{-1}\frac{12}{13}=\sin^{-1}\frac{56}{65}$.

TYPE 13 Addition and Subtraction Formulas for Inverse Trigonometric Functions

To Prove: $\cos^{-1}\frac{3}{5}+\sin^{-1}\frac{12}{13}=\sin^{-1}\frac{56}{65}$

Concept / Formula Used: $\sin^{-1}x + \sin^{-1}y = \sin^{-1}\left(x\sqrt{1-y^2} + y\sqrt{1-x^2}\right)$

Solution:

Convert $\cos^{-1}\frac{3}{5}$ to sine form: Let $A = \cos^{-1}\frac{3}{5} \implies \cos A = \frac{3}{5}$. Then $\sin A = \sqrt{1-\left(\frac{3}{5}\right)^2} = \sqrt{1-\frac{9}{25}} = \frac{4}{5}$. Thus $A = \sin^{-1}\frac{4}{5}$. Substitute into the left-hand side:

$$\text{LHS} = \sin^{-1}\frac{4}{5} + \sin^{-1}\frac{12}{13}$$

Let $x = \frac{4}{5}$ and $y = \frac{12}{13}$. Compute the square root terms:

$$\sqrt{1-x^2} = \sqrt{1-\frac{16}{25}} = \frac{3}{5}$$

$$\sqrt{1-y^2} = \sqrt{1-\frac{144}{169}} = \frac{5}{13}$$

Using $\sin^{-1}x + \sin^{-1}y = \sin^{-1}\left(x\sqrt{1-y^2} + y\sqrt{1-x^2}\right)$:

$$\begin{aligned}\text{LHS} &= \sin^{-1}\left(\frac{4}{5}\times\frac{5}{13} + \frac{12}{13}\times\frac{3}{5}\right)\\ &= \sin^{-1}\left(\frac{20}{65}+\frac{36}{65}\right)\\ &= \sin^{-1}\left(\frac{56}{65}\right) = \text{RHS}\end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

Question 19 (iv)

Prove that $\tan^{-1}\frac{1}{3} + \sec^{-1}\frac{\sqrt{5}}{2} = \frac{\pi}{4}$.

TYPE 13 **Addition and Subtraction Formulas for Inverse Trigonometric Functions**

To Prove: $\tan^{-1}\frac{1}{3} + \sec^{-1}\frac{\sqrt{5}}{2} = \frac{\pi}{4}$

Concept / Formula Used: $\sec^{-1}x = \tan^{-1}\sqrt{x^2-1} \quad x > 1 \quad \tan^{-1}a + \tan^{-1}b = \tan^{-1}\left(\frac{a+b}{1-ab}\right)$

Solution:

Let $\theta = \sec^{-1}\frac{\sqrt{5}}{2} \implies \sec\theta = \frac{\sqrt{5}}{2}$. Using $\tan\theta = \sqrt{\sec^2\theta - 1}$:

$$\begin{aligned}\tan\theta &= \sqrt{\left(\frac{\sqrt{5}}{2}\right)^2 - 1}\\ &= \sqrt{\frac{5}{4} - 1}\\ &= \sqrt{\frac{1}{4}} = \frac{1}{2}\end{aligned}$$

So $\sec^{-1}\frac{\sqrt{5}}{2} = \tan^{-1}\frac{1}{2}$. Substitute into the left-hand side:

$$\text{LHS} = \tan^{-1}\frac{1}{3} + \tan^{-1}\frac{1}{2}$$

Using $\tan^{-1}a + \tan^{-1}b = \tan^{-1}\left(\frac{a+b}{1-ab}\right)$:

$$\begin{aligned}\text{LHS} &= \tan^{-1}\left(\frac{\frac{1}{3}+\frac{1}{2}}{1-\frac{1}{3}\times\frac{1}{2}}\right)\\ &= \tan^{-1}\left(\frac{\frac{5}{6}}{1-\frac{1}{6}}\right)\\ &= \tan^{-1}\left(\frac{\frac{5}{6}}{\frac{5}{6}}\right)\\ &= \tan^{-1}(1) = \frac{\pi}{4} = \text{RHS}\end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

Question 19 (v)

Prove that $\cos^{-1}\frac{4}{5} + \tan^{-1}\frac{3}{5} = \tan^{-1}\frac{27}{11}$.

TYPE 13 Addition and Subtraction Formulas for Inverse Trigonometric Functions

To Prove: $\cos^{-1}\frac{4}{5} + \tan^{-1}\frac{3}{5} = \tan^{-1}\frac{27}{11}$

Concept / Formula Used: $\cos^{-1}x = \tan^{-1}\left(\frac{\sqrt{1-x^2}}{x}\right) \qquad x > 0$

Solution:

Convert $\cos^{-1}\frac{4}{5}$ into tangent form: Let $\theta = \cos^{-1}\frac{4}{5} \implies \cos\theta = \frac{4}{5}$. Then $\sin\theta =$

$\sqrt{1-\left(\frac{4}{5}\right)^2} = \frac{3}{5}$. Therefore $\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{3/5}{4/5} = \frac{3}{4}$. So $\cos^{-1}\frac{4}{5} = \tan^{-1}\frac{3}{4}$. Substitute into the left-hand side:

$$\text{LHS} = \tan^{-1}\frac{3}{4} + \tan^{-1}\frac{3}{5}$$

Using $\tan^{-1}a + \tan^{-1}b = \tan^{-1}\left(\frac{a+b}{1-ab}\right)$:

$$\begin{aligned}\text{LHS} &= \tan^{-1}\left(\frac{\frac{3}{4}+\frac{3}{5}}{1-\frac{3}{4}\times\frac{3}{5}}\right)\\ &= \tan^{-1}\left(\frac{\frac{15+12}{20}}{1-\frac{9}{20}}\right)\\ &= \tan^{-1}\left(\frac{\frac{27}{20}}{\frac{11}{20}}\right)\\ &= \tan^{-1}\frac{27}{11} = \text{RHS}\end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

Question 19 (vi)

Prove that $\cos^{-1}\frac{5}{\sqrt{41}} + \cot^{-1}\frac{4}{5} = \frac{\pi}{2}$.

TYPE 13 Addition and Subtraction Formulas for Inverse Trigonometric Functions

To Prove: $\cos^{-1}\frac{5}{\sqrt{41}} + \cot^{-1}\frac{4}{5} = \frac{\pi}{2}$

Concept / Formula Used: $\cos^{-1}x = \tan^{-1}\left(\frac{\sqrt{1-x^2}}{x}\right)$ $\cot^{-1}y = \tan^{-1}\left(\frac{1}{y}\right)$

Solution:

Let $A = \cos^{-1}\frac{5}{\sqrt{41}} \implies \cos A = \frac{5}{\sqrt{41}}$. Then $\sin A = \sqrt{1-\frac{25}{41}} = \sqrt{\frac{16}{41}} = \frac{4}{\sqrt{41}}$. So $\tan A = \frac{\sin A}{\cos A} = \frac{4/\sqrt{41}}{5/\sqrt{41}} = \frac{4}{5}$, giving $A = \tan^{-1}\frac{4}{5}$. Also, using $\cot^{-1}y = \tan^{-1}\left(\frac{1}{y}\right)$:

$$\cot^{-1}\frac{4}{5} = \tan^{-1}\frac{5}{4}$$

Substitute into the left-hand side:

$$\text{LHS} = \tan^{-1}\frac{4}{5} + \tan^{-1}\frac{5}{4}$$

Using the standard result $\tan^{-1} x + \tan^{-1}\left(\frac{1}{x}\right) = \frac{\pi}{2}$ for $x = \frac{4}{5}$:

$$\text{LHS} = \frac{\pi}{2} = \text{RHS}$$

Hence Proved

LHS = RHS, hence proved.

Question 19 (vii)

Prove that $\sin^{-1}\frac{1}{\sqrt{17}} + \cos^{-1}\frac{9}{\sqrt{85}} = \tan^{-1}\frac{1}{2}$.

TYPE 13 **Addition and Subtraction Formulas for Inverse Trigonometric Functions**

To Prove: $\sin^{-1}\frac{1}{\sqrt{17}} + \cos^{-1}\frac{9}{\sqrt{85}} = \tan^{-1}\frac{1}{2}$

Concept / Formula Used: $\tan^{-1} a + \tan^{-1} b = \tan^{-1}\left(\frac{a+b}{1-ab}\right)$

Solution:

Convert both terms to tangent form: For $\sin^{-1}\frac{1}{\sqrt{17}}$: $\sin A = \frac{1}{\sqrt{17}} \implies \cos A = \frac{4}{\sqrt{17}} \implies \tan A = \frac{1}{4}$. So $\sin^{-1}\frac{1}{\sqrt{17}} = \tan^{-1}\frac{1}{4}$.

For $\cos^{-1}\frac{9}{\sqrt{85}}$: $\cos B = \frac{9}{\sqrt{85}} \implies \sin B = \frac{2}{\sqrt{85}} \implies \tan B = \frac{2}{9}$. So $\cos^{-1}\frac{9}{\sqrt{85}} = \tan^{-1}\frac{2}{9}$.

Substitute both into the left-hand side:

$$\text{LHS} = \tan^{-1}\frac{1}{4} + \tan^{-1}\frac{2}{9}$$

Using $\tan^{-1} a + \tan^{-1} b = \tan^{-1}\left(\frac{a+b}{1-ab}\right)$:

$$\begin{aligned}\text{LHS} &= \tan^{-1}\left(\frac{\frac{1}{4}+\frac{2}{9}}{1-\frac{1}{4}\times\frac{2}{9}}\right)\\ &= \tan^{-1}\left(\frac{\frac{9+8}{36}}{1-\frac{2}{36}}\right)\\ &= \tan^{-1}\left(\frac{\frac{17}{36}}{\frac{34}{36}}\right)\\ &= \tan^{-1}\left(\frac{17}{34}\right)\\ &= \tan^{-1}\frac{1}{2} = \text{RHS}\end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

Question 19 (viii)

Prove that $\sin^{-1}\frac{8}{17}+\cos^{-1}\frac{4}{5}=\cot^{-1}\frac{36}{77}$.

TYPE 13 **Addition and Subtraction Formulas for Inverse Trigonometric Functions**

To Prove: $\sin^{-1}\frac{8}{17}+\cos^{-1}\frac{4}{5}=\cot^{-1}\frac{36}{77}$

Concept / Formula Used: $\tan^{-1}a+\tan^{-1}b=\tan^{-1}\left(\frac{a+b}{1-ab}\right)$ $\cot^{-1}x=\tan^{-1}\left(\frac{1}{x}\right)$

Solution:

Convert terms to tangent form: For $\sin^{-1}\frac{8}{17}$: $\sin A=\frac{8}{17}\implies\cos A=\frac{15}{17}\implies\tan A=\frac{8}{15}$. For $\cos^{-1}\frac{4}{5}$: $\cos B=\frac{4}{5}\implies\sin B=\frac{3}{5}\implies\tan B=\frac{3}{4}$.

Substitute into the left-hand side:

$$\text{LHS}=\tan^{-1}\frac{8}{15}+\tan^{-1}\frac{3}{4}$$

Using $\tan^{-1}a+\tan^{-1}b=\tan^{-1}\left(\frac{a+b}{1-ab}\right)$:

$$\begin{aligned}\text{LHS}&=\tan^{-1}\left(\frac{\frac{8}{15}+\frac{3}{4}}{1-\frac{8}{15}\times\frac{3}{4}}\right)\\&=\tan^{-1}\left(\frac{\frac{32+45}{60}}{1-\frac{24}{60}}\right)\\&=\tan^{-1}\left(\frac{\frac{77}{60}}{\frac{36}{60}}\right)\\&=\tan^{-1}\frac{77}{36}\end{aligned}$$

Using $\tan^{-1}\frac{77}{36}=\cot^{-1}\frac{36}{77}$:

$$\text{LHS}=\cot^{-1}\frac{36}{77}=\text{RHS}$$

Hence Proved

LHS = RHS, hence proved.

Question 19 (ix)

Prove that $\sin^{-1}\dfrac{12}{13}+\cos^{-1}\dfrac{4}{5}+\tan^{-1}\dfrac{63}{16}=\pi$.

TYPE 13 **Addition and Subtraction Formulas for Inverse Trigonometric Functions**

★ **Likely Exam Question**

To Prove: $\sin^{-1}\dfrac{12}{13}+\cos^{-1}\dfrac{4}{5}+\tan^{-1}\dfrac{63}{16}=\pi$

Concept / Formula Used: $x>0, y>0, xy>1$ $\quad \tan^{-1}x+\tan^{-1}y=\pi+\tan^{-1}\left(\dfrac{x+y}{1-xy}\right)$

Solution:

Convert the first two terms to tangent form: For $\sin^{-1}\dfrac{12}{13}$: $\sin A=\dfrac{12}{13} \implies \cos A=\dfrac{5}{13} \implies \tan A=\dfrac{12}{5}$. For $\cos^{-1}\dfrac{4}{5}$: $\cos B=\dfrac{4}{5} \implies \sin B=\dfrac{3}{5} \implies \tan B=\dfrac{3}{4}$.
Substitute into LHS:

$$\text{LHS}=\tan^{-1}\frac{12}{5}+\tan^{-1}\frac{3}{4}+\tan^{-1}\frac{63}{16}$$

Consider $x=\dfrac{12}{5}$ and $y=\dfrac{3}{4}$. Here $xy=\dfrac{12}{5}\times\dfrac{3}{4}=\dfrac{9}{5}>1$. Using $\tan^{-1}x+\tan^{-1}y=\pi+\tan^{-1}\left(\dfrac{x+y}{1-xy}\right)$:

$$\begin{aligned}
\tan^{-1}\frac{12}{5}+\tan^{-1}\frac{3}{4} &= \pi+\tan^{-1}\left(\frac{\frac{12}{5}+\frac{3}{4}}{1-\frac{12}{5}\times\frac{3}{4}}\right)\\
&= \pi+\tan^{-1}\left(\frac{\frac{48+15}{20}}{1-\frac{36}{20}}\right)\\
&= \pi+\tan^{-1}\left(\frac{\frac{63}{20}}{-\frac{16}{20}}\right)\\
&= \pi+\tan^{-1}\left(-\frac{63}{16}\right)\\
&= \pi-\tan^{-1}\frac{63}{16}
\end{aligned}$$

Substitute this result back into the left-hand side:

$$\begin{aligned}
\text{LHS} &= \left(\pi-\tan^{-1}\frac{63}{16}\right)+\tan^{-1}\frac{63}{16}\\
&= \pi=\text{RHS}
\end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

Question 19 (x)

Prove that $2\tan^{-1}\frac{1}{5} + \sec^{-1}\frac{5\sqrt{2}}{7} + 2\tan^{-1}\frac{1}{8} = \frac{\pi}{4}$.

TYPE 13 Addition and Subtraction Formulas for Inverse Trigonometric Functions

To Prove: $2\tan^{-1}\frac{1}{5} + \sec^{-1}\frac{5\sqrt{2}}{7} + 2\tan^{-1}\frac{1}{8} = \frac{\pi}{4}$

Concept / Formula Used: $2\tan^{-1}x = \tan^{-1}\left(\frac{2x}{1-x^2}\right)$ $\sec^{-1}y = \tan^{-1}\sqrt{y^2-1}$

Solution:

First group the terms with coefficient 2:

$$2\left(\tan^{-1}\frac{1}{5} + \tan^{-1}\frac{1}{8}\right)$$

Using $\tan^{-1}a + \tan^{-1}b = \tan^{-1}\left(\frac{a+b}{1-ab}\right)$:

$$\begin{aligned}\tan^{-1}\frac{1}{5} + \tan^{-1}\frac{1}{8} &= \tan^{-1}\left(\frac{\frac{1}{5}+\frac{1}{8}}{1-\frac{1}{5}\times\frac{1}{8}}\right)\\ &= \tan^{-1}\left(\frac{\frac{13}{40}}{\frac{39}{40}}\right)\\ &= \tan^{-1}\frac{1}{3}\end{aligned}$$

Now multiply by 2 using $2\tan^{-1}x = \tan^{-1}\left(\frac{2x}{1-x^2}\right)$:

$$2\tan^{-1}\frac{1}{3} = \tan^{-1}\left(\frac{2\times\frac{1}{3}}{1-\frac{1}{9}}\right) = \tan^{-1}\left(\frac{\frac{2}{3}}{\frac{8}{9}}\right) = \tan^{-1}\frac{3}{4}$$

Next, convert $\sec^{-1}\dfrac{5\sqrt{2}}{7}$ to tangent form: Let $\theta = \sec^{-1}\dfrac{5\sqrt{2}}{7} \implies \sec\theta = \dfrac{5\sqrt{2}}{7}$.

$$\begin{aligned}
\tan\theta &= \sqrt{\sec^2\theta - 1} \\
&= \sqrt{\left(\frac{5\sqrt{2}}{7}\right)^2 - 1} \\
&= \sqrt{\frac{50}{49} - 1} \\
&= \sqrt{\frac{1}{49}} = \frac{1}{7}
\end{aligned}$$

So $\sec^{-1}\dfrac{5\sqrt{2}}{7} = \tan^{-1}\dfrac{1}{7}$.

Now combine both parts into LHS:

$$\text{LHS} = \tan^{-1}\frac{3}{4} + \tan^{-1}\frac{1}{7}$$

$$\begin{aligned}
\text{LHS} &= \tan^{-1}\left(\frac{\frac{3}{4} + \frac{1}{7}}{1 - \frac{3}{4} \times \frac{1}{7}}\right) \\
&= \tan^{-1}\left(\frac{\frac{25}{28}}{\frac{25}{28}}\right) \\
&= \tan^{-1}(1) = \frac{\pi}{4} = \text{RHS}
\end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

Question 20

If $\cos^{-1}\left(\dfrac{x}{a}\right) + \cos^{-1}\left(\dfrac{y}{b}\right) = \alpha$, prove that $\dfrac{x^2}{a^2} - \dfrac{2xy}{ab}\cos\alpha + \dfrac{y^2}{b^2} = \sin^2\alpha$.

TYPE 13 Addition and Subtraction Formulas for Inverse Trigonometric Functions

★ Likely Exam Question

To Prove: $\dfrac{x^2}{a^2} - \dfrac{2xy}{ab}\cos\alpha + \dfrac{y^2}{b^2} = \sin^2\alpha$ given $\cos^{-1}\left(\dfrac{x}{a}\right) + \cos^{-1}\left(\dfrac{y}{b}\right) = \alpha$

Concept / Formula Used: $\cos^{-1}u + \cos^{-1}v = \cos^{-1}\left(uv - \sqrt{1-u^2}\sqrt{1-v^2}\right)$

Solution:

Given equation:

$$\cos^{-1}\left(\frac{x}{a}\right) + \cos^{-1}\left(\frac{y}{b}\right) = \alpha$$

Using the inverse cosine addition identity:

$$\cos^{-1}\left(\frac{x}{a}\cdot\frac{y}{b}-\sqrt{1-\frac{x^2}{a^2}}\sqrt{1-\frac{y^2}{b^2}}\right)=\alpha$$

Taking cosine on both sides:

$$\frac{xy}{ab}-\sqrt{1-\frac{x^2}{a^2}}\sqrt{1-\frac{y^2}{b^2}}=\cos\alpha$$

Isolating the square root term on the right-hand side:

$$\frac{xy}{ab}-\cos\alpha=\sqrt{1-\frac{x^2}{a^2}}\sqrt{1-\frac{y^2}{b^2}}$$

Squaring both sides:

$$\left(\frac{xy}{ab}-\cos\alpha\right)^2=\left(1-\frac{x^2}{a^2}\right)\left(1-\frac{y^2}{b^2}\right)$$

Expanding both sides:

$$\frac{x^2y^2}{a^2b^2}-\frac{2xy}{ab}\cos\alpha+\cos^2\alpha=1-\frac{y^2}{b^2}-\frac{x^2}{a^2}+\frac{x^2y^2}{a^2b^2}$$

Subtracting $\frac{x^2y^2}{a^2b^2}$ from both sides:

$$-\frac{2xy}{ab}\cos\alpha+\cos^2\alpha=1-\frac{y^2}{b^2}-\frac{x^2}{a^2}$$

Rearranging terms:

$$\frac{x^2}{a^2}-\frac{2xy}{ab}\cos\alpha+\frac{y^2}{b^2}=1-\cos^2\alpha$$

Using $1-\cos^2\alpha=\sin^2\alpha$:

$$\frac{x^2}{a^2}-\frac{2xy}{ab}\cos\alpha+\frac{y^2}{b^2}=\sin^2\alpha$$

Hence Proved

LHS = RHS, hence proved.

Question 21 (i)

Evaluate $\tan\left(2\tan^{-1}\frac{1}{2}-\cot^{-1}3\right)$.

TYPE 9 **Evaluating Trigonometric Functions of Inverse Arguments**

To Find: The value of $\tan\left(2\tan^{-1}\frac{1}{2}-\cot^{-1}3\right)$

Concept / Formula Used: $2\tan^{-1} x = \tan^{-1}\left(\dfrac{2x}{1-x^2}\right)$ $\tan(A-B) = \dfrac{\tan A - \tan B}{1+\tan A \tan B}$

Solution:

First simplify $2\tan^{-1}\dfrac{1}{2}$:

$$\begin{aligned}2\tan^{-1}\frac{1}{2} &= \tan^{-1}\left(\frac{2\times\frac{1}{2}}{1-\left(\frac{1}{2}\right)^2}\right)\\ &= \tan^{-1}\left(\frac{1}{1-\frac{1}{4}}\right)\\ &= \tan^{-1}\left(\frac{1}{\frac{3}{4}}\right)\\ &= \tan^{-1}\frac{4}{3}\end{aligned}$$

Convert $\cot^{-1} 3$ to tangent form:

$$\cot^{-1} 3 = \tan^{-1}\frac{1}{3}$$

Let $A = 2\tan^{-1}\dfrac{1}{2} = \tan^{-1}\dfrac{4}{3} \implies \tan A = \dfrac{4}{3}$. Let $B = \cot^{-1} 3 = \tan^{-1}\dfrac{1}{3} \implies \tan B = \dfrac{1}{3}$.

Evaluating $\tan(A-B)$:

$$\begin{aligned}\tan(A-B) &= \frac{\tan A - \tan B}{1+\tan A\tan B}\\ &= \frac{\frac{4}{3}-\frac{1}{3}}{1+\frac{4}{3}\times\frac{1}{3}}\\ &= \frac{1}{1+\frac{4}{9}}\\ &= \frac{1}{\frac{13}{9}}\\ &= \frac{9}{13}\end{aligned}$$

Final Answer

$$\tan\left(2\tan^{-1}\frac{1}{2} - \cot^{-1} 3\right) = \frac{9}{13}$$

Question 21 (ii)

Evaluate $\tan\left(2\tan^{-1}\dfrac{1}{5} - \dfrac{\pi}{4}\right)$.

TYPE 9 Evaluating Trigonometric Functions of Inverse Arguments

To Find: The value of $\tan\left(2\tan^{-1}\dfrac{1}{5}-\dfrac{\pi}{4}\right)$

Concept / Formula Used: $2\tan^{-1}x = \tan^{-1}\left(\dfrac{2x}{1-x^2}\right)$ $\tan(A-B) = \dfrac{\tan A-\tan B}{1+\tan A\tan B}$

Solution:

First simplify $2\tan^{-1}\dfrac{1}{5}$:

$$\begin{aligned}
2\tan^{-1}\frac{1}{5} &= \tan^{-1}\left(\frac{2\times\frac{1}{5}}{1-\left(\frac{1}{5}\right)^2}\right)\\
&= \tan^{-1}\left(\frac{\frac{2}{5}}{1-\frac{1}{25}}\right)\\
&= \tan^{-1}\left(\frac{\frac{2}{5}}{\frac{24}{25}}\right)\\
&= \tan^{-1}\left(\frac{2}{5}\times\frac{25}{24}\right)\\
&= \tan^{-1}\frac{5}{12}
\end{aligned}$$

Let $A = 2\tan^{-1}\dfrac{1}{5} = \tan^{-1}\dfrac{5}{12} \implies \tan A = \dfrac{5}{12}$. Let $B = \dfrac{\pi}{4} \implies \tan B = 1$.

Evaluating $\tan\left(A-\dfrac{\pi}{4}\right)$:

$$\begin{aligned}
\tan\left(A-\frac{\pi}{4}\right) &= \frac{\tan A-1}{1+\tan A}\\
&= \frac{\frac{5}{12}-1}{1+\frac{5}{12}}\\
&= \frac{-\frac{7}{12}}{\frac{17}{12}}\\
&= -\frac{7}{17}
\end{aligned}$$

Final Answer

$\tan\left(2\tan^{-1}\dfrac{1}{5}-\dfrac{\pi}{4}\right) = -\dfrac{7}{17}$

Question 22 (i)

Find the value of $\tan\left(\cos^{-1}\frac{3}{5}+\tan^{-1}\frac{1}{4}\right)$.

TYPE 9 **Evaluating Trigonometric Functions of Inverse Arguments**

To Find: The value of $\tan\left(\cos^{-1}\frac{3}{5}+\tan^{-1}\frac{1}{4}\right)$

Concept / Formula Used: $\cos^{-1}x = \tan^{-1}\left(\frac{\sqrt{1-x^2}}{x}\right)$ $x > 0$ $\tan(A+B) = \frac{\tan A+\tan B}{1-\tan A\tan B}$

Solution:

Convert $\cos^{-1}\frac{3}{5}$ into tangent form: Let $A = \cos^{-1}\frac{3}{5} \implies \cos A = \frac{3}{5}$. Then $\sin A = \sqrt{1-\frac{9}{25}} = \frac{4}{5} \implies \tan A = \frac{4}{3}$.

Let $B = \tan^{-1}\frac{1}{4} \implies \tan B = \frac{1}{4}$.

Evaluating $\tan(A+B)$:

$$\begin{aligned}\tan(A+B) &= \frac{\tan A+\tan B}{1-\tan A\tan B}\\ &= \frac{\frac{4}{3}+\frac{1}{4}}{1-\frac{4}{3}\times\frac{1}{4}}\\ &= \frac{\frac{16+3}{12}}{1-\frac{4}{12}}\\ &= \frac{\frac{19}{12}}{\frac{8}{12}}\\ &= \frac{19}{8}\end{aligned}$$

Final Answer

$\tan\left(\cos^{-1}\frac{3}{5}+\tan^{-1}\frac{1}{4}\right) = \frac{19}{8}$

Question 22 (ii)

Find the value of $\cos\left(\sin^{-1}\frac{3}{5}+\sin^{-1}\frac{5}{13}\right)$.

TYPE 9 **Evaluating Trigonometric Functions of Inverse Arguments**

To Find: The value of $\cos\left(\sin^{-1}\frac{3}{5}+\sin^{-1}\frac{5}{13}\right)$

Concept / Formula Used: $\cos(A+B)=\cos A\cos B-\sin A\sin B$

Solution:

Let $A=\sin^{-1}\frac{3}{5} \implies \sin A=\frac{3}{5}$. Then $\cos A=\sqrt{1-\left(\frac{3}{5}\right)^2}=\sqrt{1-\frac{9}{25}}=\frac{4}{5}$.

Let $B=\sin^{-1}\frac{5}{13} \implies \sin B=\frac{5}{13}$. Then $\cos B=\sqrt{1-\left(\frac{5}{13}\right)^2}=\sqrt{1-\frac{25}{169}}=\frac{12}{13}$.

Applying the expansion formula for $\cos(A+B)$:

$$\begin{aligned}\cos(A+B)&=\cos A\cos B-\sin A\sin B\\&=\frac{4}{5}\times\frac{12}{13}-\frac{3}{5}\times\frac{5}{13}\\&=\frac{48}{65}-\frac{15}{65}\\&=\frac{33}{65}\end{aligned}$$

Final Answer

$\cos\left(\sin^{-1}\frac{3}{5}+\sin^{-1}\frac{5}{13}\right)=\frac{33}{65}$

Question 22 (iii)

Find the value of $\cot\left(\cos^{-1}\frac{4}{5}+\sin^{-1}\frac{2}{\sqrt{13}}\right)$.

TYPE 9 Evaluating Trigonometric Functions of Inverse Arguments

To Find: The value of $\cot\left(\cos^{-1}\frac{4}{5}+\sin^{-1}\frac{2}{\sqrt{13}}\right)$

Concept / Formula Used: $\tan(A+B)=\dfrac{\tan A+\tan B}{1-\tan A\tan B}$ $\qquad \cot\theta=\dfrac{1}{\tan\theta}$

Solution:

Let $A=\cos^{-1}\frac{4}{5} \implies \cos A=\frac{4}{5}, \sin A=\frac{3}{5} \implies \tan A=\frac{3}{4}$.

Let $B=\sin^{-1}\frac{2}{\sqrt{13}} \implies \sin B=\frac{2}{\sqrt{13}}, \cos B=\sqrt{1-\frac{4}{13}}=\frac{3}{\sqrt{13}} \implies \tan B=\frac{2}{3}$.

Calculate $\tan(A+B)$:

$$\begin{aligned}\tan(A+B) &= \frac{\tan A + \tan B}{1 - \tan A \tan B} \\ &= \frac{\frac{3}{4}+\frac{2}{3}}{1-\frac{3}{4}\times\frac{2}{3}} \\ &= \frac{\frac{9+8}{12}}{1-\frac{6}{12}} \\ &= \frac{\frac{17}{12}}{\frac{6}{12}} \\ &= \frac{17}{6}\end{aligned}$$

Using $\cot(A+B) = \dfrac{1}{\tan(A+B)}$:

$$\cot(A+B) = \frac{6}{17}$$

Final Answer

$\cot\left(\cos^{-1}\frac{4}{5} + \sin^{-1}\frac{2}{\sqrt{13}}\right) = \frac{6}{17}$

Question 22 (iv)

Find the value of $\sin\left(2\tan^{-1}\frac{2}{3}\right) + \cos\left(\tan^{-1}\sqrt{3}\right)$.

TYPE 9 Evaluating Trigonometric Functions of Inverse Arguments

★ Likely Exam Question

To Find: The value of $\sin\left(2\tan^{-1}\frac{2}{3}\right) + \cos\left(\tan^{-1}\sqrt{3}\right)$

Concept / Formula Used: $\sin(2\theta) = \dfrac{2\tan\theta}{1+\tan^2\theta}$ $\qquad \cos(\tan^{-1}x) = \dfrac{1}{\sqrt{1+x^2}}$

Solution:

Evaluate the first term $\sin\left(2\tan^{-1}\frac{2}{3}\right)$: Let $\theta = \tan^{-1}\frac{2}{3} \implies \tan\theta = \frac{2}{3}$. Using

$\sin(2\theta) = \dfrac{2\tan\theta}{1+\tan^2\theta}$:

$$\begin{aligned}\sin\left(2\tan^{-1}\frac{2}{3}\right) &= \frac{2\times\frac{2}{3}}{1+\left(\frac{2}{3}\right)^2} \\ &= \frac{\frac{4}{3}}{1+\frac{4}{9}} \\ &= \frac{\frac{4}{3}}{\frac{13}{9}} \\ &= \frac{4}{3}\times\frac{9}{13} \\ &= \frac{12}{13}\end{aligned}$$

Evaluate the second term $\cos\left(\tan^{-1}\sqrt{3}\right)$: Since $\tan^{-1}\sqrt{3} = \dfrac{\pi}{3}$:

$$\cos\left(\tan^{-1}\sqrt{3}\right) = \cos\frac{\pi}{3} = \frac{1}{2}$$

Summing both values:

$$\begin{aligned}\text{Total Value} &= \frac{12}{13}+\frac{1}{2} \\ &= \frac{24+13}{26} \\ &= \frac{37}{26}\end{aligned}$$

Final Answer

$\sin\left(2\tan^{-1}\dfrac{2}{3}\right) + \cos\left(\tan^{-1}\sqrt{3}\right) = \dfrac{37}{26}$

Question 23 (i)

Solve the equation for x: $\tan^{-1}\left(\dfrac{2-x}{2+x}\right) = \dfrac{1}{2}\tan^{-1}\dfrac{x}{2}, x > 0.$

TYPE 12 Solving Inverse Trigonometric Equations

To Find: The value of $x > 0$ satisfying $\tan^{-1}\left(\dfrac{2-x}{2+x}\right) = \dfrac{1}{2}\tan^{-1}\dfrac{x}{2}$

Concept / Formula Used: $\tan^{-1}\left(\dfrac{a-b}{a+b}\right) = \tan^{-1}1 - \tan^{-1}\left(\dfrac{b}{a}\right) = \dfrac{\pi}{4} - \tan^{-1}\left(\dfrac{b}{a}\right)$

Solution:

Rewrite the argument on the left-hand side:

$$\tan^{-1}\left(\frac{2-x}{2+x}\right) = \tan^{-1}\left(\frac{1-\frac{x}{2}}{1+\frac{x}{2}}\right)$$

Using $\tan^{-1}\left(\frac{1-y}{1+y}\right) = \tan^{-1} 1 - \tan^{-1} y = \frac{\pi}{4} - \tan^{-1} y$ for $y = \frac{x}{2}$:

$$\tan^{-1}\left(\frac{2-x}{2+x}\right) = \frac{\pi}{4} - \tan^{-1}\frac{x}{2}$$

Substitute into the given equation:

$$\frac{\pi}{4} - \tan^{-1}\frac{x}{2} = \frac{1}{2}\tan^{-1}\frac{x}{2}$$

Add $\tan^{-1}\frac{x}{2}$ to both sides:

$$\frac{\pi}{4} = \frac{1}{2}\tan^{-1}\frac{x}{2} + \tan^{-1}\frac{x}{2}$$

$$\frac{\pi}{4} = \frac{3}{2}\tan^{-1}\frac{x}{2}$$

Multiply both sides by $\frac{2}{3}$:

$$\tan^{-1}\frac{x}{2} = \frac{\pi}{6}$$

Taking tangent on both sides:

$$\frac{x}{2} = \tan\frac{\pi}{6}$$

$$\frac{x}{2} = \frac{1}{\sqrt{3}}$$

Multiply by 2:

$$x = \frac{2}{\sqrt{3}}$$

Since $x = \frac{2}{\sqrt{3}} > 0$, it satisfies the given domain condition.

Final Answer

$x = \dfrac{2}{\sqrt{3}}$

Question 23 (ii)

Solve the equation for x: $\tan^{-1} 4x + \tan^{-1} 6x = \frac{\pi}{4}$.

TYPE 12 **Solving Inverse Trigonometric Equations**

★ **Likely Exam Question**

To Find: The value of x satisfying $\tan^{-1} 4x + \tan^{-1} 6x = \frac{\pi}{4}$

Concept / Formula Used: $\tan^{-1} a + \tan^{-1} b = \tan^{-1}\left(\frac{a+b}{1-ab}\right)$

Solution:

Applying the addition identity:

$$\tan^{-1}\left(\frac{4x+6x}{1-(4x)(6x)}\right) = \frac{\pi}{4}$$

$$\tan^{-1}\left(\frac{10x}{1-24x^2}\right) = \frac{\pi}{4}$$

Taking tangent on both sides:

$$\frac{10x}{1-24x^2} = \tan\frac{\pi}{4} = 1$$

Cross-multiplying:

$$10x = 1 - 24x^2$$

Rearranging into standard quadratic form:

$$24x^2 + 10x - 1 = 0$$

Factorizing the quadratic expression:

$$\begin{aligned} 24x^2 + 12x - 2x - 1 &= 0 \\ 12x(2x+1) - 1(2x+1) &= 0 \\ (12x-1)(2x+1) &= 0 \end{aligned}$$

This gives two possible roots:

$$x = \frac{1}{12} \quad \text{or} \quad x = -\frac{1}{2}$$

Check validity: For $x = -\frac{1}{2}$, both $4x = -2 < 0$ and $6x = -3 < 0$, making $\tan^{-1} 4x + \tan^{-1} 6x < 0 \neq \frac{\pi}{4}$. Thus $x = -\frac{1}{2}$ is an extraneous root. For $x = \frac{1}{12}$, both $4x > 0$ and $6x > 0$ with product $(4x)(6x) = \frac{1}{6} < 1$, which satisfies the equation.

Final Answer

$x = \frac{1}{12}$

Common Mistake: Forgetting to verify roots in inverse trig equations: $x = -\frac{1}{2}$ makes both arguments negative, which produces a negative sum and must be rejected.

Question 23 (iii)

Solve the equation for x: $\tan^{-1}(x+1) + \tan^{-1}(x-1) = \tan^{-1}\dfrac{6}{17}$.

TYPE 12 **Solving Inverse Trigonometric Equations**

To Find: The value of x satisfying $\tan^{-1}(x+1) + \tan^{-1}(x-1) = \tan^{-1}\dfrac{6}{17}$

Concept / Formula Used: $\tan^{-1} a + \tan^{-1} b = \tan^{-1}\left(\dfrac{a+b}{1-ab}\right)$

Solution:

Apply the identity $\tan^{-1} a + \tan^{-1} b = \tan^{-1}\left(\dfrac{a+b}{1-ab}\right)$ for $a = x+1$ and $b = x-1$:

$$\tan^{-1}\left(\frac{(x+1)+(x-1)}{1-(x+1)(x-1)}\right) = \tan^{-1}\frac{6}{17}$$

Simplifying the argument on the left-hand side:

$$\tan^{-1}\left(\frac{2x}{1-(x^2-1)}\right) = \tan^{-1}\frac{6}{17}$$

$$\tan^{-1}\left(\frac{2x}{2-x^2}\right) = \tan^{-1}\frac{6}{17}$$

Equating the arguments:

$$\frac{2x}{2-x^2} = \frac{6}{17}$$

Divide numerator of both sides by 2:

$$\frac{x}{2-x^2} = \frac{3}{17}$$

Cross-multiplying:

$$\begin{aligned} 17x &= 3(2-x^2) \\ 17x &= 6-3x^2 \\ 3x^2+17x-6 &= 0 \end{aligned}$$

Factorizing:

$$\begin{aligned} 3x^2+18x-x-6 &= 0 \\ 3x(x+6)-1(x+6) &= 0 \\ (3x-1)(x+6) &= 0 \end{aligned}$$

This gives:

$$x = \frac{1}{3} \quad \text{or} \quad x = -6$$

Checking $x = -6$: LHS $= \tan^{-1}(-5) + \tan^{-1}(-7) < 0$, while RHS $= \tan^{-1}\dfrac{6}{17} > 0$, so $x = -6$ is extraneous. For $x = \dfrac{1}{3}$, LHS matches RHS.

Final Answer

$x = \dfrac{1}{3}$

Question 23 (iv)

Solve the equation for x: $\tan^{-1}\left(\dfrac{x-2}{x-1}\right) + \tan^{-1}\left(\dfrac{x+2}{x+1}\right) = \dfrac{\pi}{4}$.

TYPE 12 **Solving Inverse Trigonometric Equations**

★ **Likely Exam Question**

To Find: The value of x satisfying $\tan^{-1}\left(\dfrac{x-2}{x-1}\right) + \tan^{-1}\left(\dfrac{x+2}{x+1}\right) = \dfrac{\pi}{4}$

Concept / Formula Used: $\tan^{-1} a + \tan^{-1} b = \tan^{-1}\left(\dfrac{a+b}{1-ab}\right)$

Solution:

Applying $\tan^{-1} a + \tan^{-1} b = \tan^{-1}\left(\dfrac{a+b}{1-ab}\right)$:

$$\tan^{-1}\left(\frac{\frac{x-2}{x-1} + \frac{x+2}{x+1}}{1 - \frac{x-2}{x-1}\cdot\frac{x+2}{x+1}}\right) = \frac{\pi}{4}$$

Taking tangent on both sides:

$$\frac{\frac{(x-2)(x+1)+(x+2)(x-1)}{(x-1)(x+1)}}{\frac{(x-1)(x+1)-(x-2)(x+2)}{(x-1)(x+1)}} = \tan\frac{\pi}{4} = 1$$

Cancelling the common denominator $(x-1)(x+1)$:

$$\frac{(x-2)(x+1)+(x+2)(x-1)}{(x-1)(x+1)-(x-2)(x+2)} = 1$$

Expand numerator and denominator:

$$\text{Numerator} = (x^2 - x - 2) + (x^2 + x - 2) = 2x^2 - 4$$
$$\text{Denominator} = (x^2 - 1) - (x^2 - 4) = x^2 - 1 - x^2 + 4 = 3$$

Substitute these expressions back:

$$\frac{2x^2 - 4}{3} = 1$$

Multiply by 3:

$$\begin{aligned} 2x^2 - 4 &= 3 \\ 2x^2 &= 7 \\ x^2 &= \frac{7}{2} \\ x &= \pm\sqrt{\frac{7}{2}} \end{aligned}$$

Final Answer

$x = \pm\sqrt{\frac{7}{2}}$

Question 23 (v)

Solve the equation for x: $\tan^{-1}\left(\frac{x-2}{x-3}\right) + \tan^{-1}\left(\frac{x+2}{x+3}\right) = \frac{\pi}{4}$.

TYPE 12 **Solving Inverse Trigonometric Equations**

To Find: The value of x satisfying $\tan^{-1}\left(\frac{x-2}{x-3}\right) + \tan^{-1}\left(\frac{x+2}{x+3}\right) = \frac{\pi}{4}$

Concept / Formula Used: $\tan^{-1} a + \tan^{-1} b = \tan^{-1}\left(\frac{a+b}{1-ab}\right)$

Solution:

Applying the addition identity:

$$\tan^{-1}\left(\frac{\frac{x-2}{x-3} + \frac{x+2}{x+3}}{1 - \frac{x-2}{x-3} \cdot \frac{x+2}{x+3}}\right) = \frac{\pi}{4}$$

Taking tangent on both sides:

$$\frac{(x-2)(x+3) + (x+2)(x-3)}{(x-3)(x+3) - (x-2)(x+2)} = \tan\frac{\pi}{4} = 1$$

Expand numerator and denominator:

$$\text{Numerator} = (x^2 + x - 6) + (x^2 - x - 6) = 2x^2 - 12$$
$$\text{Denominator} = (x^2 - 9) - (x^2 - 4) = x^2 - 9 - x^2 + 4 = -5$$

Substitute back:

$$\frac{2x^2 - 12}{-5} = 1$$

Multiply by -5:

$$\begin{aligned} 2x^2 - 12 &= -5 \\ 2x^2 &= 7 \\ x^2 &= \frac{7}{2} \\ x &= \pm\sqrt{\frac{7}{2}} \end{aligned}$$

Final Answer

$x = \pm\sqrt{\frac{7}{2}}$

Question 23 (vi)

Solve the equation for x: $\tan^{-1}\left(\frac{x-3}{x-4}\right) + \tan^{-1}\left(\frac{x+3}{x+4}\right) = \frac{\pi}{4}$.

TYPE 12 **Solving Inverse Trigonometric Equations**

To Find: The value of x satisfying $\tan^{-1}\left(\frac{x-3}{x-4}\right) + \tan^{-1}\left(\frac{x+3}{x+4}\right) = \frac{\pi}{4}$

Concept / Formula Used: $\tan^{-1} a + \tan^{-1} b = \tan^{-1}\left(\frac{a+b}{1-ab}\right)$

Solution:

Applying the addition formula:

$$\tan^{-1}\left(\frac{\frac{x-3}{x-4} + \frac{x+3}{x+4}}{1 - \frac{x-3}{x-4}\cdot\frac{x+3}{x+4}}\right) = \frac{\pi}{4}$$

Taking tangent on both sides:

$$\frac{(x-3)(x+4) + (x+3)(x-4)}{(x-4)(x+4) - (x-3)(x+3)} = \tan\frac{\pi}{4} = 1$$

Expand numerator and denominator:

$$\text{Numerator} = (x^2 + x - 12) + (x^2 - x - 12) = 2x^2 - 24$$
$$\text{Denominator} = (x^2 - 16) - (x^2 - 9) = x^2 - 16 - x^2 + 9 = -7$$

Substitute back:

$$\frac{2x^2 - 24}{-7} = 1$$

Multiply by -7:

$$\begin{aligned} 2x^2 - 24 &= -7 \\ 2x^2 &= 17 \\ x^2 &= \frac{17}{2} \\ x &= \pm\sqrt{\frac{17}{2}} \end{aligned}$$

Final Answer

$x = \pm\sqrt{\frac{17}{2}}$

Question 23 (vii)

Solve the equation for x: $\cos(\sin^{-1} x) = \frac{1}{9}$.

TYPE 12 **Solving Inverse Trigonometric Equations**

To Find: The value of x satisfying $\cos(\sin^{-1} x) = \frac{1}{9}$

Concept / Formula Used: $\cos(\sin^{-1} x) = \sqrt{1 - x^2}$

Solution:

Let $\theta = \sin^{-1} x \implies \sin\theta = x$. Using the standard identity $\cos\theta = \sqrt{1 - \sin^2\theta}$:

$$\cos(\sin^{-1} x) = \sqrt{1 - x^2}$$

Given equation becomes:

$$\sqrt{1 - x^2} = \frac{1}{9}$$

Squaring both sides:

$$\begin{aligned} 1 - x^2 &= \frac{1}{81} \\ x^2 &= 1 - \frac{1}{81} \\ x^2 &= \frac{80}{81} \end{aligned}$$

Taking square root:

$$\begin{aligned} x &= \pm\frac{\sqrt{80}}{9} \\ &= \pm\frac{4\sqrt{5}}{9} \end{aligned}$$

Final Answer

$x = \pm\frac{4\sqrt{5}}{9}$

Question 23 (viii)

Solve the equation for x: $\cos(2\sin^{-1}x) = \frac{1}{9}$.

TYPE 12 **Solving Inverse Trigonometric Equations**

To Find: The value of x satisfying $\cos(2\sin^{-1}x) = \frac{1}{9}$

Concept / Formula Used: $\cos(2\theta) = 1 - 2\sin^2\theta$

Solution:

Let $\theta = \sin^{-1}x \implies \sin\theta = x$. The equation becomes:

$$\cos(2\theta) = \frac{1}{9}$$

Using the double angle identity $\cos(2\theta) = 1 - 2\sin^2\theta$:

$$1 - 2\sin^2\theta = \frac{1}{9}$$

Substitute $\sin\theta = x$:

$$1 - 2x^2 = \frac{1}{9}$$

Rearranging terms:

$$\begin{aligned} 2x^2 &= 1 - \frac{1}{9} \\ 2x^2 &= \frac{8}{9} \\ x^2 &= \frac{4}{9} \\ x &= \pm\frac{2}{3} \end{aligned}$$

Final Answer

$x = \pm\frac{2}{3}$

Question 23 (ix)

Solve the equation for x: $\cos^{-1}(\sin(\cos^{-1}x)) = \frac{\pi}{6}$.

TYPE 12 **Solving Inverse Trigonometric Equations**

To Find: The value of x satisfying $\cos^{-1}(\sin(\cos^{-1} x)) = \frac{\pi}{6}$

Concept / Formula Used: $\sin(\cos^{-1} x) = \sqrt{1-x^2}$

Solution:

Given equation:

$$\cos^{-1}(\sin(\cos^{-1} x)) = \frac{\pi}{6}$$

Taking cosine on both sides:

$$\sin(\cos^{-1} x) = \cos\frac{\pi}{6}$$

Since $\cos\frac{\pi}{6} = \frac{\sqrt{3}}{2}$:

$$\sin(\cos^{-1} x) = \frac{\sqrt{3}}{2}$$

Using the standard identity $\sin(\cos^{-1} x) = \sqrt{1-x^2}$:

$$\sqrt{1-x^2} = \frac{\sqrt{3}}{2}$$

Squaring both sides:

$$1 - x^2 = \frac{3}{4}$$
$$x^2 = 1 - \frac{3}{4}$$
$$x^2 = \frac{1}{4}$$
$$x = \pm\frac{1}{2}$$

Final Answer

$x = \pm\frac{1}{2}$

Question 23 (x)

Solve the equation for x: $\sin^{-1}(\cos(\sin^{-1} x)) = \frac{\pi}{3}$.

TYPE 12 **Solving Inverse Trigonometric Equations**

To Find: The value of x satisfying $\sin^{-1}(\cos(\sin^{-1} x)) = \frac{\pi}{3}$

Concept / Formula Used: $\cos(\sin^{-1} x) = \sqrt{1-x^2}$

Solution:

Given equation:

$$\sin^{-1}(\cos(\sin^{-1} x)) = \frac{\pi}{3}$$

Taking sine on both sides:

$$\cos(\sin^{-1} x) = \sin\frac{\pi}{3}$$

Since $\sin\frac{\pi}{3} = \frac{\sqrt{3}}{2}$:

$$\cos(\sin^{-1} x) = \frac{\sqrt{3}}{2}$$

Using the standard identity $\cos(\sin^{-1} x) = \sqrt{1-x^2}$:

$$\sqrt{1-x^2} = \frac{\sqrt{3}}{2}$$

Squaring both sides:

$$\begin{aligned} 1 - x^2 &= \frac{3}{4} \\ x^2 &= 1 - \frac{3}{4} \\ x^2 &= \frac{1}{4} \\ x &= \pm\frac{1}{2} \end{aligned}$$

Final Answer

$x = \pm\frac{1}{2}$

Question 23 (xi)

Solve the equation for x: $\sin^{-1}\frac{8}{x} + \sin^{-1}\frac{15}{x} = \frac{\pi}{2}$.

TYPE 12 Solving Inverse Trigonometric Equations

★ Likely Exam Question

To Find: The value of x satisfying $\sin^{-1}\frac{8}{x} + \sin^{-1}\frac{15}{x} = \frac{\pi}{2}$

Concept / Formula Used: $\sin^{-1} A + \cos^{-1} A = \frac{\pi}{2}$

Solution:

Given equation:

$$\sin^{-1}\frac{8}{x} + \sin^{-1}\frac{15}{x} = \frac{\pi}{2}$$

Transpose $\sin^{-1}\dfrac{15}{x}$ to the right-hand side:

$$\sin^{-1}\frac{8}{x} = \frac{\pi}{2} - \sin^{-1}\frac{15}{x}$$

Using the co-function identity $\dfrac{\pi}{2} - \sin^{-1} y = \cos^{-1} y$:

$$\sin^{-1}\frac{8}{x} = \cos^{-1}\frac{15}{x}$$

Taking sine on both sides:

$$\frac{8}{x} = \sin\left(\cos^{-1}\frac{15}{x}\right)$$

Using $\sin(\cos^{-1} y) = \sqrt{1-y^2}$:

$$\frac{8}{x} = \sqrt{1 - \left(\frac{15}{x}\right)^2}$$

Squaring both sides:

$$\begin{aligned} \frac{64}{x^2} &= 1 - \frac{225}{x^2} \\ \frac{64}{x^2} + \frac{225}{x^2} &= 1 \\ \frac{289}{x^2} &= 1 \\ x^2 &= 289 \\ x &= \pm 17 \end{aligned}$$

Since $\sin^{-1}\dfrac{8}{x}$ and $\sin^{-1}\dfrac{15}{x}$ must add up to positive $\dfrac{\pi}{2}$, x must be positive. Therefore $x = 17$.

Final Answer

$x = 17$

Common Mistake: Taking $x = -17$ leads to negative angles on the left side, summing to $-\frac{\pi}{2}$, which contradicts the right side $\frac{\pi}{2}$. Hence $x = -17$ must be discarded.

Question 23 (xii)

Solve the equation for x: $\sin^{-1}\dfrac{2a}{1+a^2} + \sin^{-1}\dfrac{2b}{1+b^2} = 2\tan^{-1} x$.

TYPE 12 Solving Inverse Trigonometric Equations

★ Likely Exam Question

To Find: The value of x satisfying $\sin^{-1}\dfrac{2a}{1+a^2} + \sin^{-1}\dfrac{2b}{1+b^2} = 2\tan^{-1} x$

Concept / Formula Used: $\sin^{-1}\left(\frac{2y}{1+y^2}\right) = 2\tan^{-1} y \qquad |y| \leq 1$

Solution:

Using the standard identity $\sin^{-1}\left(\frac{2y}{1+y^2}\right) = 2\tan^{-1} y$:

$$\sin^{-1}\frac{2a}{1+a^2} = 2\tan^{-1} a$$

$$\sin^{-1}\frac{2b}{1+b^2} = 2\tan^{-1} b$$

Substitute these into the given equation:

$$2\tan^{-1} a + 2\tan^{-1} b = 2\tan^{-1} x$$

Dividing the entire equation by 2:

$$\tan^{-1} a + \tan^{-1} b = \tan^{-1} x$$

Using the addition identity $\tan^{-1} a + \tan^{-1} b = \tan^{-1}\left(\frac{a+b}{1-ab}\right)$:

$$\tan^{-1}\left(\frac{a+b}{1-ab}\right) = \tan^{-1} x$$

Taking tangent on both sides:

$$x = \frac{a+b}{1-ab}$$

Final Answer

$x = \frac{a+b}{1-ab}$